



CSN305: Theory of Computation

Session: 2025-26

B. Tech (CSE) 6th Semester

Unit-1: Introduction to Finite Automata

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Topics to be covered



Unit 1: Introduction to Finite Automata.

(8 L)

Introduction to Mathematical foundation for automata: Mathematical preliminaries, alphabets, strings, languages, states, transition, transition graph, generalized transition graph.

Finite Automata: Deterministic Finite Automata, Non-Deterministic Finite Automata, Non-Deterministic Finite Automata with ϵ transitions, minimization of DFA.

Set

Set

- A **set** is a collection of objects.
- The objects in a set are called **elements** of the set.



- **Examples:**

1. $A = \{11, 12, 21, 22\}$

2. $B = \{11, 12, 21, 11, 12, 22\}$

3. $C = \{x \mid x \text{ is odd integer greater than } 1\}$

4. $D = \{x \mid x \in B \text{ and } x \leq 11\}$

} Roster Notation

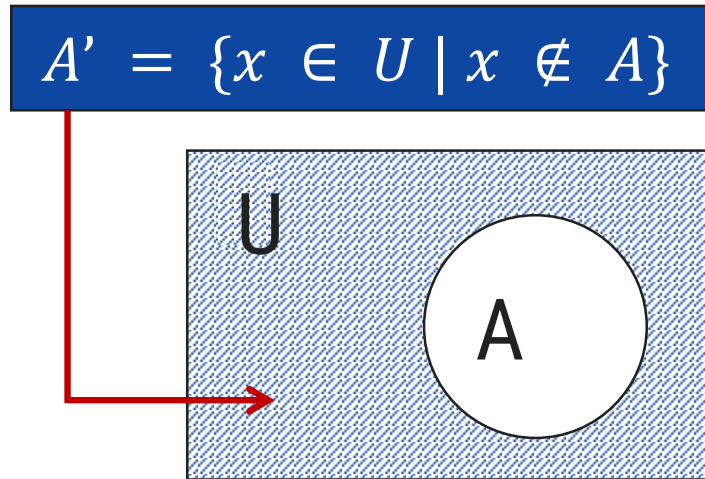
} Set-builder Notation

Operations on Sets

- Operations on the sets are:

1. Complement
2. Union
3. Intersection
4. Set Difference
5. Symmetric Difference
6. Cartesian product

- ▶ The **complement** of a set A is the set A' of everything that is not an element of A from Universal Set U .



- ▶ Example:
 $U = \{1,2,3,4,5\}$
 $A = \{1,2\}$
 $A' = \{3,4,5\}$

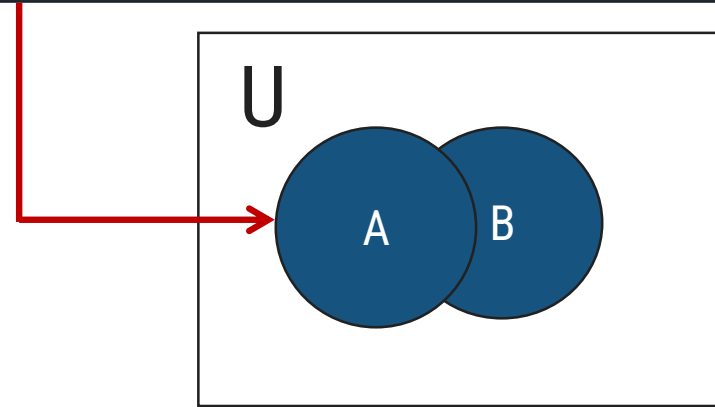
Operations on Sets

- Operations on the sets are:

1. Complement
2. Union
3. Intersection
4. Set Difference
5. Symmetric Difference
6. Cartesian product

- ▶ The **Union** ($A \cup B$) is a collection of all distinct elements from both the set A and B.

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$



- ▶ Example:

$$A = \{1, 3, 5, 7, 9\}$$

$$B = \{1, 2, 3, 4, 5\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 7, 9\}$$

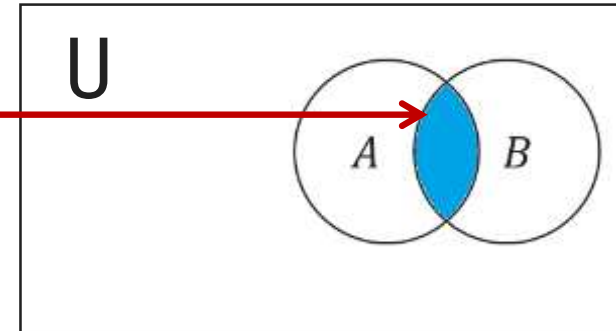
Operations on Sets

- Operations on the sets are:

1. Complement
2. Union
3. Intersection
4. Set Difference
5. Symmetric Difference
6. Cartesian product

- ▶ The **intersection** $A \cap B$ of two sets A and B is the set that contains all elements of A that also belong to B , but no other elements.

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$



- ▶ Example:

$$A = \{1, 3, 5, 7, 9\}$$

$$B = \{1, 2, 3, 4, 5\}$$

$$A \cap B = \{1, 3, 5\}$$

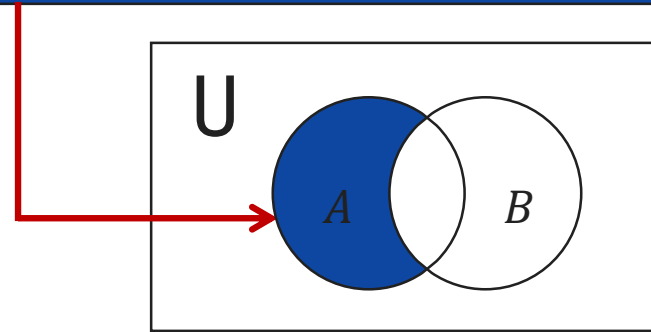
Operations on Sets

- Operations on the sets are:

1. Complement
2. Union
3. Intersection
4. Set Difference
5. Symmetric Difference
6. Cartesian product

- ▶ The **set difference** $A - B$ of two sets A and B is the set of everything in A but not in B .

$$\begin{aligned} A - B &= \{x \mid x \in A \text{ and } x \notin B\} \\ &= \{x \mid x \in A\} \cap \{x \mid x \notin B\} \\ &= A \cap B' \end{aligned}$$



- ▶ Example:

$$A = \{1, 3, 5, 7, 9\}$$

$$B = \{1, 2, 3, 4, 5\}$$

$$A - B = \{7, 9\}$$

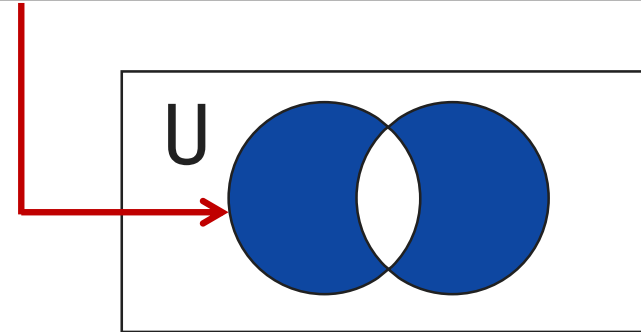
Operations on Sets

- Operations on the sets are:

1. Complement
2. Union
3. Intersection
4. Set Difference
5. Symmetric Difference
6. Cartesian product

- ▶ The **symmetric difference** $A \ominus B$ of two sets A and B is the set of everything in A but not in B or the set of everything in B but not in A .

$$A \ominus B = (A - B) \cup (B - A)$$



- ▶ Example:

$$A = \{1, 3, 5, 7, 9\}$$

$$B = \{1, 2, 3, 4, 5\}$$

$$A \ominus B = \{7, 9, 2, 4\}$$

Operations on Sets

- Operations on the sets are:

1. Complement
2. Union
3. Intersection
4. Set Difference
5. Symmetric Difference
6. Cartesian product

- ▶ The **Cartesian product** $A \times B$ of two sets A and B is the set of all **ordered pairs** (a, b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

- ▶ Example:

$$A = \{1, 3, 5\}$$

$$B = \{2, 4\}$$

$$A \times B = \{(1,2), (1,4), (3,2), (3,4), (5,2), (5,4)\}$$

Set of identities

- Commutative laws

$$\begin{aligned}A \cap B &= B \cap A \\A \cup B &= B \cup A\end{aligned}$$

- Associative laws

$$\begin{aligned}A \cap (B \cap C) &= (A \cap B) \cap C \\A \cup (B \cup C) &= (A \cup B) \cup C\end{aligned}$$

- Distributive laws

$$\begin{aligned}A \cup (B \cap C) &= (A \cup B) \cap (A \cup C) \\A \cap (B \cup C) &= (A \cap B) \cup (A \cap C)\end{aligned}$$

Set of identities

- Idempotent laws

$$\begin{aligned}A \cup A &= A \\A \cap A &= A\end{aligned}$$

- Absorptive laws

$$\begin{aligned}A \cup (A \cap B) &= A \\A \cap (A \cup B) &= A\end{aligned}$$

- De Morgan laws

$$\begin{aligned}(A \cup B)' &= A' \cap B' \\(A \cap B)' &= A' \cup B'\end{aligned}$$

Set of identities

- Other complements laws

$$\begin{aligned}(A')' &= A \\ A \cap A' &= \Phi \\ A \cup A' &= U\end{aligned}$$

- Other empty set laws

$$\begin{aligned}A \cup \Phi &= A \\ A \cap \Phi &= \Phi\end{aligned}$$

- Other universal set laws

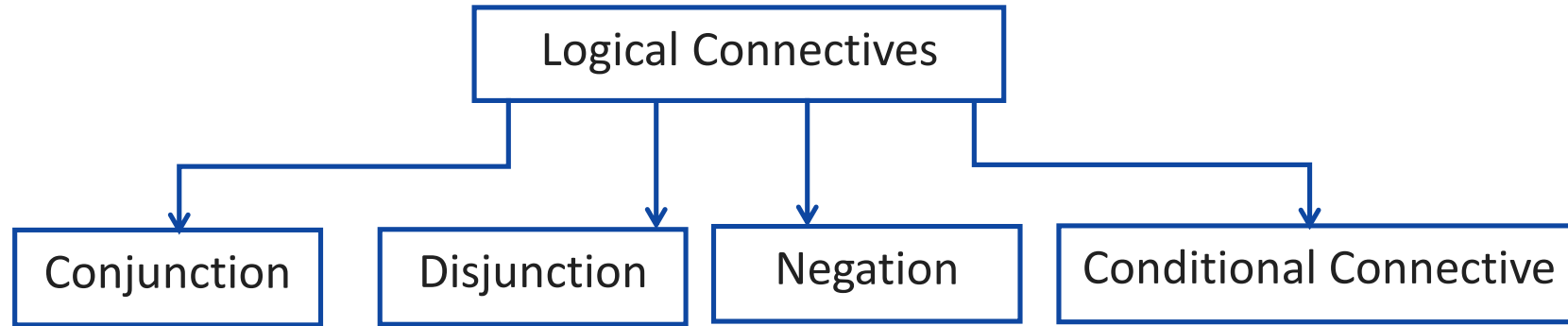
$$\begin{aligned}A \cup U &= U \\ A \cap U &= A\end{aligned}$$

Logic

Propositions

- Declarative statement that is sufficiently **objective, meaningful** and precise to **have a truth value (true or false)** is known as **proposition**.
- **Examples:**
 1. p : Fourteen is an even integer.
 2. r : $0 = 0$
 3. q : Mumbai is the capital city of India.
 4. s : $a^2 + b^2 = 4$

Logical Connectives



-
- The logical connective **Conjunction** (And) is **true** only when **both** of the propositions are **true**.

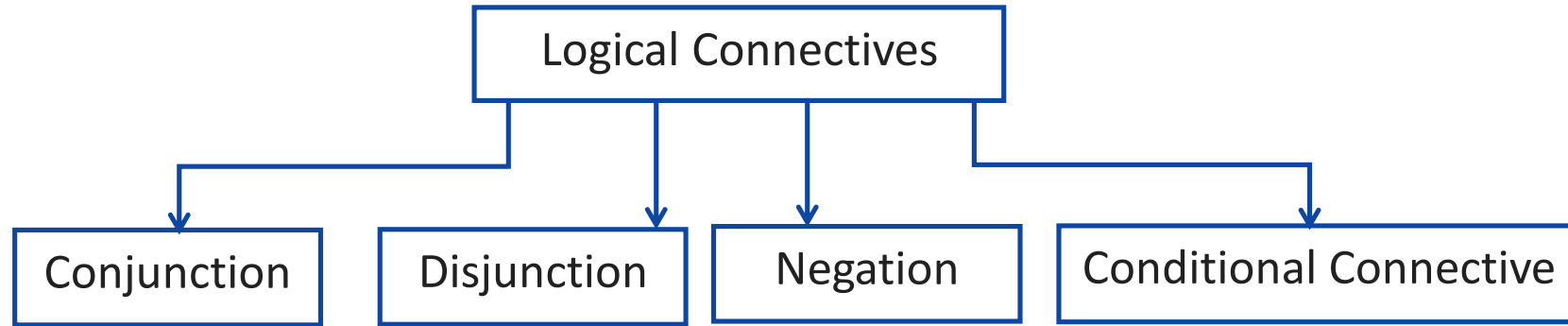
- **Example:**

Truth table

- p : It is raining
- q : It is warm
- r : It is raining **AND** it is warm

p	q	$r = p \wedge q$

Logical Connectives



-
- The logical **disjunction**, or logical OR, is **true** if **one or both** of the propositions are **true**.

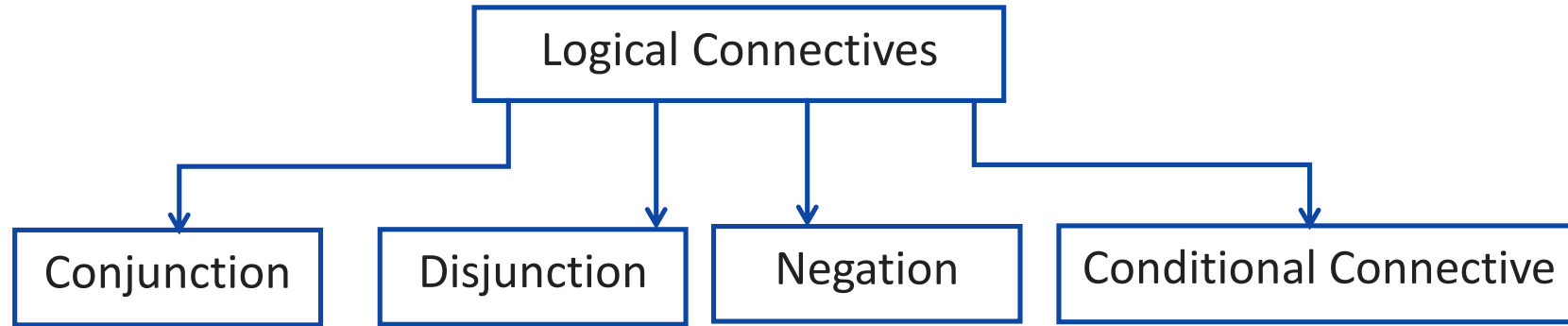
- **Example:**

Truth table

- $p : 2 + 2 = 5$
- $q : 1 < 2$
- $r : 2 + 2 = 5$ **OR** $1 < 2$

p	q	$r = p \vee q$

Logical Connectives



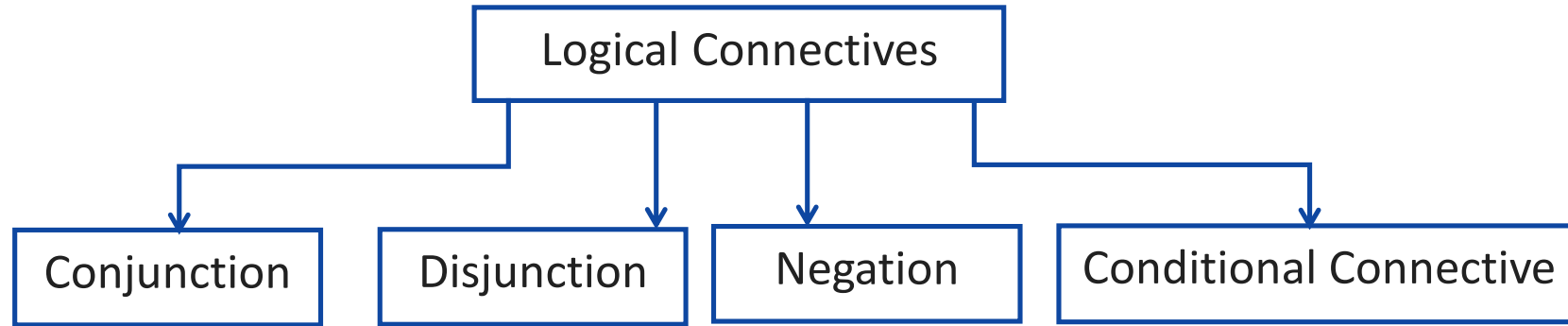
-
- $\neg p$, the **negation** of a proposition p , is also a proposition.

- **Example:** **Truth table**

- p : John studies.
- $\neg p$: John does **NOT** study.

P	$\neg P$

Logical Connectives



- The proposition $p \rightarrow q$ is commonly read as “if p then q ”.
- **Example:** $P \rightarrow$ I will win the lottery. $Q \rightarrow$ I will buy car for you.

I win the lottery	I will buy car for you	Promise kept/ broken

p	q	$p \rightarrow q$

Logical Connectives

- The statement $p \rightarrow q$ can be read as following:
 1. “**if** p then q”
 2. “q **if** p”
 3. “p **only if** q”
- Consider following two statements:
 1. “p if q” ($q \rightarrow p$)
 2. “p only if q” ($p \rightarrow q$)
- If we make conjunction of (1) & (2) then,
- $(p \rightarrow q) \wedge (q \rightarrow p) = p \leftrightarrow q$ (biconditional) “p only if q, and p if q”
- Often read as “**p if and only if q**”

Tautology and Contradiction

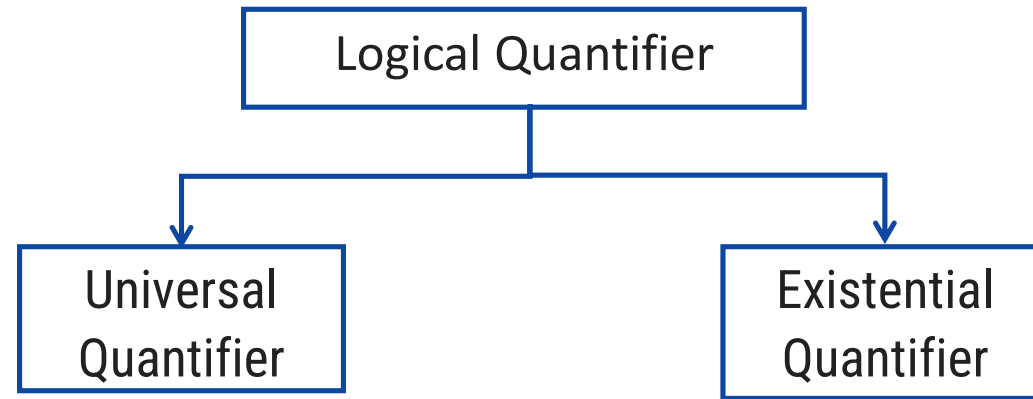
- A Compound proposition is called tautology if it is **true in every case**.
- **Example:**

P	$\neg P$	$P \vee \neg P$

Tautology

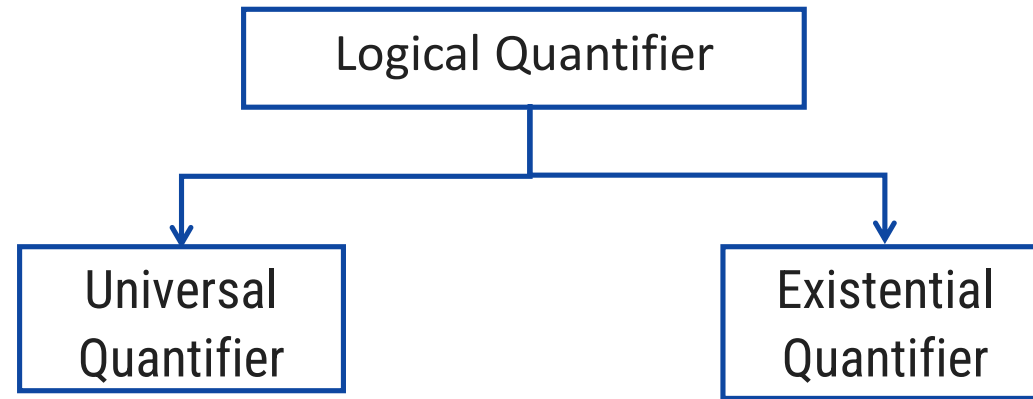
- A contradiction is opposite.
- If p is tautology, $\neg P$ is contradiction.

Logical Quantifiers



-
- Represented by an upside-down A: $\forall$ ("for all").
 - **Example:**
 - Let $P(x) = x+1 > x$, $\forall x P(x)$
 - English translation: "for all values of x , $P(x)$ is true"
 - English translation: "for all values of x , $x+1 > x$ is true"

Logical Quantifiers



-
- Represented by $\exists$: $\exists$ ("for exists").
 - Example:
 - Let $P(x) = x+1 > x$
 - There is a numerical value for which $x+1 > x$
 - Thus, $\exists x P(x)$ is true

Functions

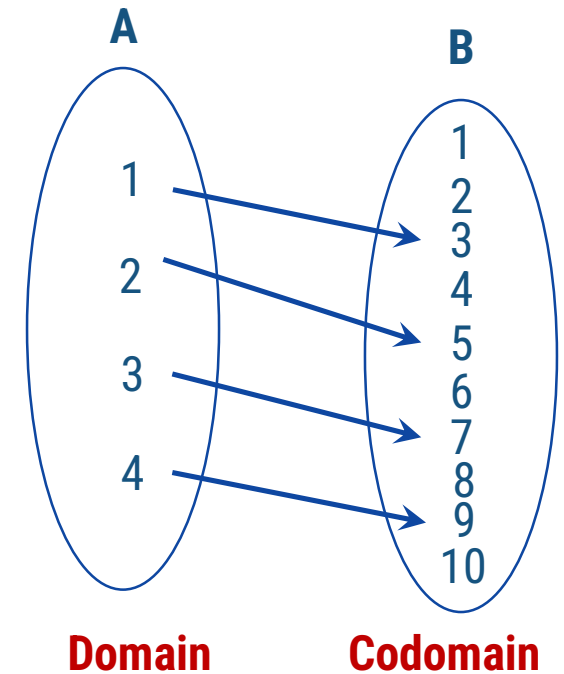
Functions

- **Domain:** What can go into the function is called domain.
- **Codomain:** What may possibly come out from a function is codomain.
- **Range:** What actually come out from a function is range. The range of function is subset of codomain
- **Example:**

$$f: N \rightarrow N, f(x) = 2x + 1$$

$$\begin{aligned} f(1) &= 2(1) + 1 = 3 \\ f(2) &= 2(2) + 1 = 5 \\ f(3) &= 2(3) + 1 = 7 \\ f(4) &= 2(4) + 1 = 9 \end{aligned}$$

Range



- The range of function $f(x) = \{3, 5, 7, 9\}$

Onto Function

- If the **range** of function and **codomain** of function **are equal** or every element of the codomain is actually one of the values of the function, then function is said to be **onto** or **surjective** or **surjection**.

- Example: $f : A \rightarrow B, f(x) = x^2$ where,

$A = \{-2, -1, 1, 2, 3, 4\}$ and $B = \{1, 4, 9, 16\}$

$$f(-2) = 4$$

$$f(-1) = 1$$

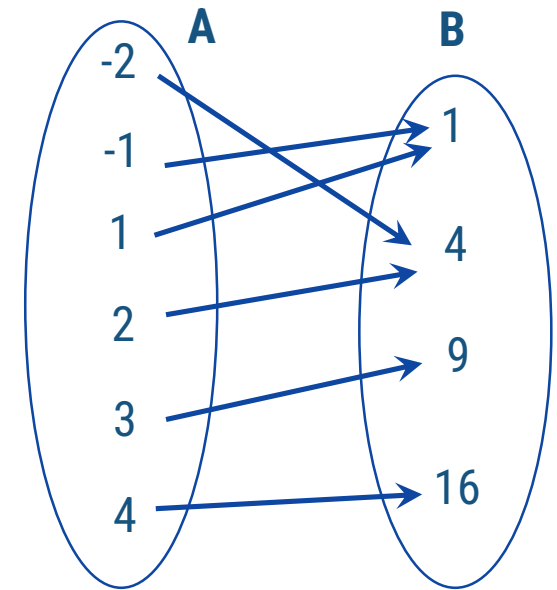
$$f(1) = 1$$

$$f(2) = 4$$

$$f(3) = 9$$

$$f(4) = 16$$

- The range of function $f(A) = \{1, 4, 9, 16\} = B$



One-to-One Function

- A function for which **every element of the range** of the function **corresponds to exactly one element of the domain** is known as **One-to-One** or **injective** or **injection**.

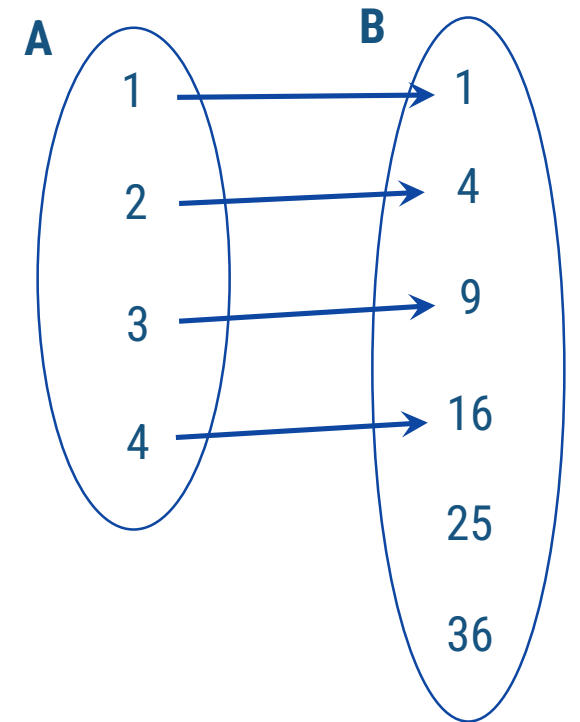
- Example: $f : A \rightarrow B, f(x) = x^2$ where,
 $A = \{1,2,3,4\}$ and $B = \{1,4,9,16,25,36\}$

$$f(1) = 1$$

$$f(2) = 4$$

$$f(3) = 9$$

$$f(4) = 16$$



Bijection Function

- If function is both **one-to-one** and **onto** then function is called **Bijection function**.
- Example: $f : A \rightarrow B, f(x) = x^2$ where,

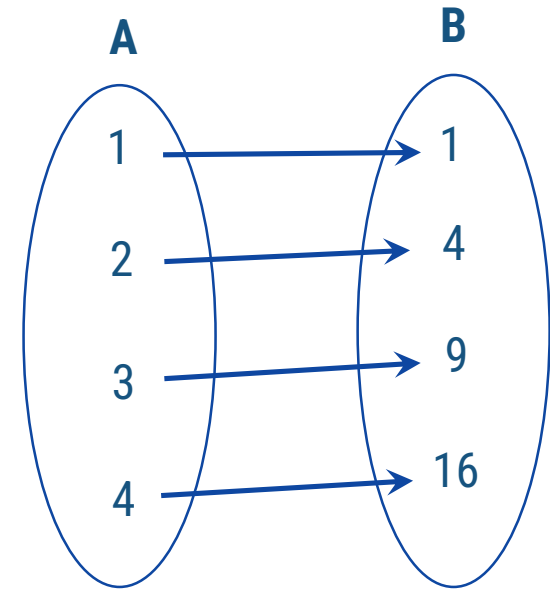
$A = \{1,2,3,4\}$ and $B = \{1,4,9,16\}$

$$f(1) = 1$$

$$f(2) = 4$$

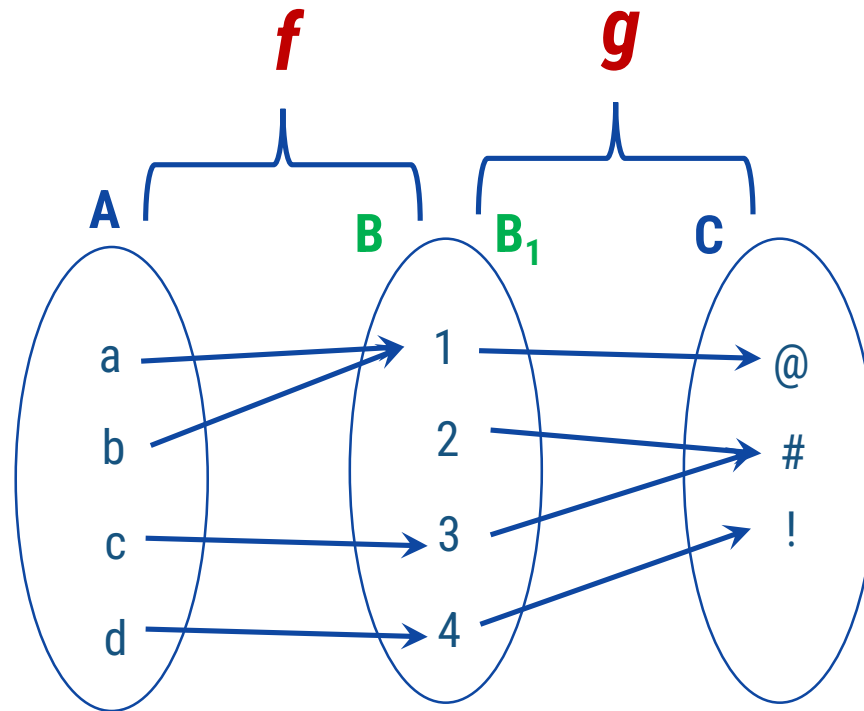
$$f(3) = 9$$

$$f(4) = 16$$



Compositions of Function

- Let $f : A \rightarrow B$ and $g : B_1 \rightarrow C$, the **range of f is a subset of B_1** , then $g(f(x))$ makes sense for each $x \in A$ and the function **$h : A \rightarrow C$** defined by **$h(x) = g(f(x))$** is called the composition of g and f .
- It is written as **$h = g \circ f$**
- Example:**
 - $h(c) = g(f(x)) = \#$

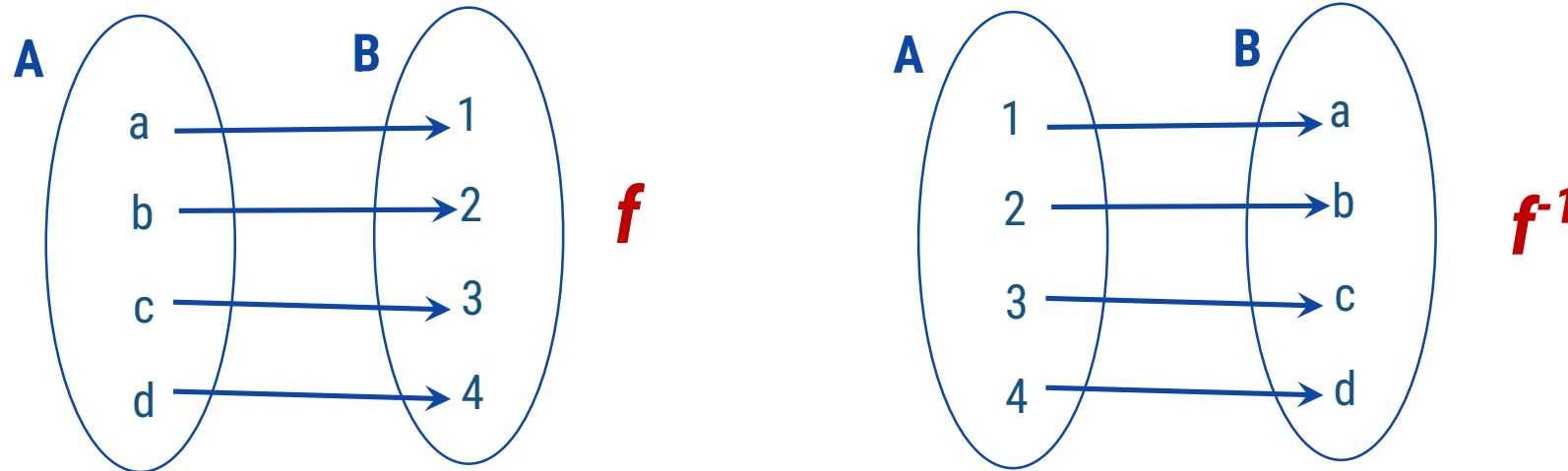


Inverse of Function

- Let f be a function whose domain is the set X , and whose range is the set Y . Then f is **invertible** if there exists a function g with domain Y and range X , with the property:

$$f(x) = y \Leftrightarrow g(y) = x$$

- To be invertible a function **must be** both an **injection** and a **surjection**.
- Example:



Relations

Relations

- A **relation** on a set A is defined as subset of $A \times A$.
- The relation R is denoted as **aRb** where $a, b \in A$ and pair $(a, b) \in R$.
- **Example:**

$$N = \{1, 2, 3\}$$

$$N \times N = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$$

- The **'='** relation on $N \times N$ is : $\{(1,1), (2,2), (3,3)\}$

where

$$1 = 1$$

$$2 = 2$$

$$3 = 3$$

Properties of Equivalence Relations

- Assume that R is a relation on a set A , in other words, $R \subseteq A \times A$, where $(x, y) \in R$ to indicate x is related to y via Relation R .
 1. R is **reflexive** if for every $x \in A$, xRx
 2. R is **symmetric** if for every x and y in A , if xRy , then yRx
 3. R is **transitive** if for every x, y and z in A , if xRy and yRz , then xRz .
 4. R is an **equivalence** relation on A , if R is **reflexive**, **symmetric** and **transitive**.

Example: Equivalence Relation

- $A=\{a, b\}$, $R=\{(a, a), (b, b), (a, b), (b, a)\}$
 - Reflexive: $\{(a, a), (b, b)\}$ ✓
 - Symmetric: $\{(a, b), (b, a)\}$ ✓
 - Transitive: $\{(a, a), (a, b), (b, a), (b, b), (a, b), (b, a), (a, a), (b, b), (b, a), (a, a), (b, a)\}$ ✓
 - $(a, a), (a, b), (b, a), (b, b)$
 - $(a, b), (b, b), (a, b)$
 - $(a, b), (b, a), (a, a)$
 - $(b, a), (a, b), (b, b)$
 - $(b, a), (a, a), (b, a)$
- Above relation is **Equivalence relation** because it is Reflexive, symmetric and transitive.

Exercise

1. $A=\{1, 2, 3\}$, $R=\{(1, 2), (1, 1), (2, 1), (2, 2), (3, 2), (3, 3)\}$ is equivalent relation?
2. $A=\{1, 2, 3, 4\}$, $R=\{(1, 1), (2, 2), (2, 3), (3, 2), (4, 2), (4, 4)\}$ is equivalent relation?
3. $A=\{0, 1, 2\}$, $R=\{(0, 0), (1, 1), (2, 2), (1, 0), (2, 1)\}$ is equivalent relation?

Proof

Proof

- A proof of a statement is essentially just a **convincing argument** that **the statement is true**.
- A typical step in proof is to derive some statement from:
 1. Assumptions or hypotheses
 2. Statements that have already been derived
 3. Other generally accepted facts
- There are several methods for establishing a proof, some of them are:
 1. Direct proof
 2. By contradiction
 3. By mathematical induction

Rational & Irrational numbers

- A rational number is a number that can be in the form **m/n** where **m** and **n** are integers and **n** is not equal to zero.

- Examples:

$$1.5 = 3/2$$

$$6 = 12/2$$

$$5 = 15/3$$

Rational Numbers

~~$\pi = 22/7$~~
**Irrational
Numbers**

3.14159265 3589793238462643383279502884197.....

Prove: $\sqrt{2}$ is Irrational

- **Definition:** A real number is rational if there are two integers m and n so that $x = m/n$.
- **Proof:**
- Suppose for the sake of contradiction that $\sqrt{2}$ is rational.
- Then there exists some integers m' and n' such that $\sqrt{2} = m'/n'$.
- By dividing both m' and n' by all the factors that are common to both, we obtain $\sqrt{2} = m/n$, for some integers m and n having no common factors.
- Since $\sqrt{2} = m/n$, $m = n\sqrt{2}$. Squaring both sides of this equation, we obtain $m^2 = 2n^2$, and therefore m^2 is even.
- If a and b are odd, then ab is odd. Since a conditional statement is logically equivalent to its contra positive, we may conclude that for any a and b , if ab is not odd, then either a is not odd or b is not odd.

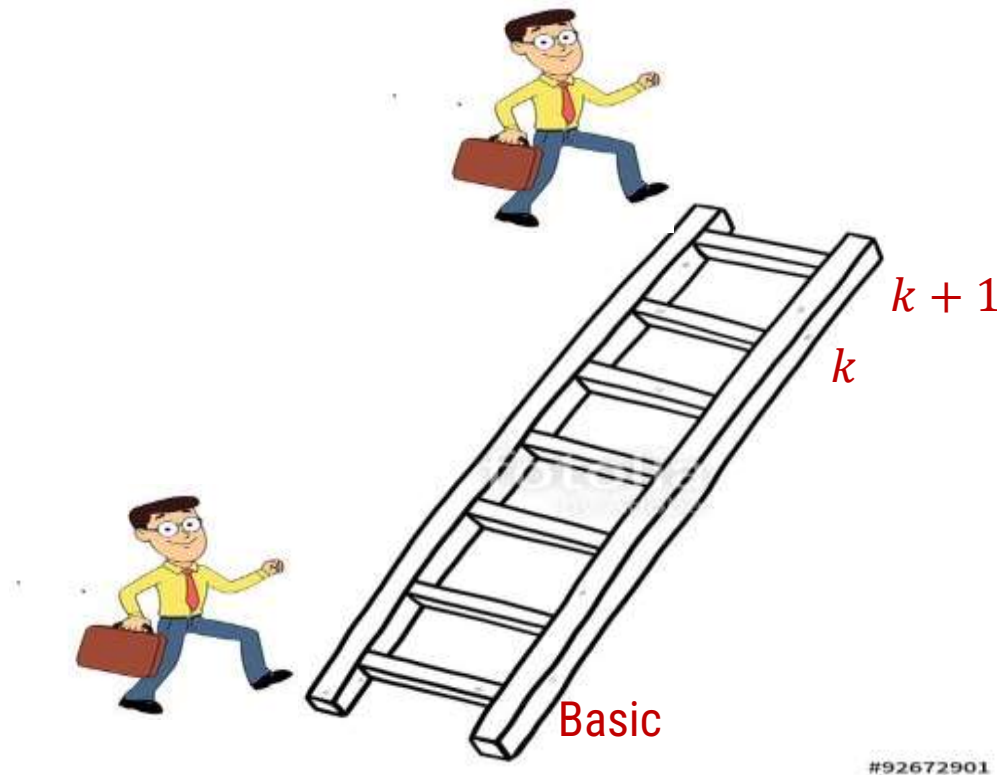
Prove: $\sqrt{2}$ is Irrational

- However, an integer is not odd if and only if it is even, and so for any a and b , if ab is even, a or b is even.
- If we apply this when $a = b = m$, we conclude that since m^2 is even, m must be even.
- This means that for some k , $m = 2k$. Therefore, $(2k)^2 = 2n^2$.
- Simplifying and cancelling 2 from both sides, we obtain $2k^2 = n^2$. Therefore n^2 is even and therefore $n = 2j$ for some j .
- We have shown that m and n are both divisible by 2. This contradicts the previous statement that m and n have no common factor.
- The assumption that $\sqrt{2}$ is rational therefore leads to a contradiction, and the conclusion is that $\sqrt{2}$ is irrational.

Principle of Mathematical Induction

Principle of Mathematical Induction

- Suppose $P(n)$ is a statement involving an integer n . Then to prove that $P(n)$ is true for every $n \geq n_0$, it is sufficient to show these two things:
 1. $P(n_0)$ is true.
 2. For any $k \geq n_0$, if $P(k)$ is true, then $P(k + 1)$ is true.



Prove $\sum_{i=1}^n i = n(n+1)/2$ using PMI

Step-1: Basic step

We must show that $P(1)$ is true.

$$P(1) = 1 \text{ (L.H.S)}$$

$$P(1) = 1(1+1)/2 = 1, \text{ And this is obviously true.}$$

Step-2: Induction Hypothesis

$$k \geq 1 \text{ and } 1 + 2 + 3 + \cdots + k = \mathbf{k(k+1)/2}$$

Step-3: Proof of Induction

$$P(k+1) = 1 + 2 + 3 + \cdots + k + (k+1)$$

$$= \mathbf{k(k+1)/2} + (k+1) \text{ (by induction hypothesis)}$$

$$= (k+1)\left(\frac{k}{2} + 1\right)$$

$$= (k+1)(k+2)/2$$

$$= (k+1)((k+1)+1)/2 \text{ (Hence Proved)}$$

Prove $1 + 3 + 5 + \dots + 2n-1 = n^2$ using PMI, $n \geq 1$

Step-1: Basic step

We must show that $P(1)$ is true.

$$P(1) = 2(1)-1 = 1 \text{ (L.H.S)}$$

$$P(1) = (1)^2 = 1 \text{ (R.H.S)}$$

And, this is obviously true.

Step-2: Induction Hypothesis

$k \geq 1$ and

$$p(k) = 1+3+5+\dots+(2k-1) = k^2$$

Step-3: Proof of Induction

$$\begin{aligned} P(k+1) &= 1+3+5+\dots+(2k-1)+(2(k+1)-1) \\ &= k^2 + (2(k+1)-1) \\ &= k^2 + (2k+2-1) \\ &= k^2 + 2k+1 \\ &= (k+1)^2 \quad (\text{Hence Proved}) \end{aligned}$$

Prove $7+13+19+\dots+(6n+1)=n(3n+4)$ using PMI, $n \geq 1$

Step-1: Basic step

We must show that $p(1)$ is true.

$$P(1) = 6n+1 = (6(1)+1) = 7$$

$$P(1) = n(3n+4) = 1(3(1)+4) = 7$$

And, this is obviously true.

Step-2: Induction Hypothesis

$k \geq 1$ and

$$p(k) = 7+13+19+\dots+(6k+1) = k(3k+4)$$

Step-3: Proof of Induction

$$\begin{aligned} P(k+1) &= 7+13+\dots+(6k+1)+(6(k+1)+1) \\ &= k(3k+4)+(6(k+1)+1) \\ &= k(3k+4)+(6k+6+1) \\ &= 3k^2+4k+6k+7 \\ &= 3k^2+10k+7 \\ &= 3k^2+3k+7k+7 \\ &= 3k(k+1)+7(k+1) \\ &= (k+1)(3k+7) \\ &= (k+1)(3k+3+4) \\ &= (k+1)(3(k+1)+4) \text{ (Hence Proved)} \end{aligned}$$

Prove $\sum_1^n \frac{1}{n(n+1)} = \frac{n}{n+1}$ using PMI

Step-1: Basic step

We must show that $p(1)$ is true.

$$P(1) = \frac{1}{1(1+1)} = \frac{1}{1(2)} = \frac{1}{2}$$

$$P(1) = \frac{1}{1+1} = \frac{1}{2}$$

And, this is obviously true.

Step-2: Induction Hypothesis

$k \geq 1$ and

$$p(k) = \frac{1}{2} + \frac{1}{6} \dots + \frac{1}{k(k+1)} =$$

$$\frac{k}{k+1}$$

Step-3: Proof of Induction

$$\begin{aligned} p(k+1) &= \frac{1}{2} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+1+1)} \\ &= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k(k+2)+1}{(k+1)(k+2)} \\ &= (k^2 + 2k + 1) / (k+1)(k+2) \\ &= \frac{(k+1)^2}{(k+1)(k+2)} \\ &= \frac{(k+1)(k+1)}{(k+1)(k+2)} \\ &= \frac{k+1}{k+2} \\ &= \frac{k+1}{(k+1)+1} \end{aligned}$$

Alphabets and Strings

Alphabets and Strings

$$\Sigma = \{a, b\}$$

- We will use small alphabets:

- Strings
 - a
 - ab
 - $abba$
 - $baba$
 - $aaabbbbaabab$

$$u = ab$$

$$v = bbbbaaa$$

$$w = abba$$

String Operations: Concatenation

$$w = a_1a_2 \cdots a_n$$

abba

$$v = b_1b_2 \cdots b_m$$

bbbbaaa

$$wv = a_1a_2 \cdots a_nb_1b_2 \cdots b_m$$

abbabbbbaaa

Reverse

$$w = a_1 a_2 \cdots a_n$$

ababaaaabbb

$$w^R = a_n \cdots a_2 a_1$$

bbbbaaababa

String Length

$$w = a_1 a_2 \cdots a_n$$

- Length:

$$|w| = n$$

- Examples:

$$|abba| = 4$$

$$|aa| = 2$$

$$|a| = 1$$

Length of Concatenation

$$|uv| = |u| + |v|$$

- Example:

$$u = aab, \quad |u| = 3$$

$$v = abaab, \quad |v| = 5$$

$$|uv| = |aababaab| = 8$$

$$|uv| = |u| + |v| = 3 + 5 = 8$$

Empty String

- A string with no letters: ε or λ

$$|\lambda| = 0$$

- Observations:

$$\lambda w = w \lambda = w$$

$$\lambda abba = abba \lambda = abba$$

Substring

- Substring of string: a subsequence of consecutive characters

•	String	Substring
	<i>abbab</i>	<i>ab</i>
	<i>abbab</i>	<i>abba</i>
	<i>abbab</i>	<i>b</i>
	<i>abbab</i>	<i>bbab</i>

Prefix and Suffix

abbab

Prefixes

λ

a

ab

abb

abba

abbab

Suffixes

abbab

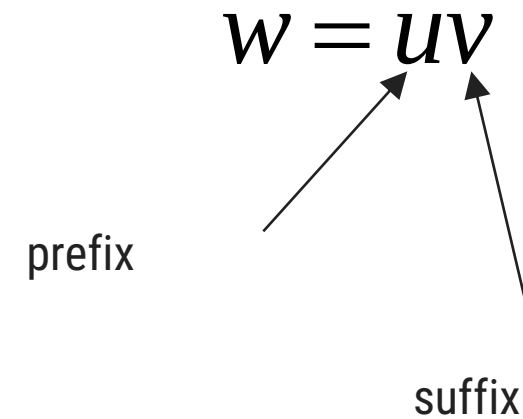
bbab

bab

ab

b

λ



Another Operation

- Definition:

$$w^n = \underbrace{ww \cdots w}_n$$

- Example:

$$(abba)^2 = abbaabba$$

- Definition:

$$w^0 = \lambda$$

- Example:

$$(abba)^0 = \lambda$$

The * Operation

Σ^* : the set of all possible strings from
alphabet Σ

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \dots\}$$

The + Operation

Σ^+ : the set of all possible strings from
alphabet Σ except λ

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \dots\}$$

$$\Sigma^+ = \Sigma^* - \lambda$$

$$\Sigma^+ = \{a, b, aa, ab, ba, bb, aaa, aab, \dots\}$$

Language

Language

- A set of strings all of which are chosen from some Σ^* , where Σ is a particular alphabet, is called a language. If Σ is an alphabet, and $L \subseteq \Sigma^*$, then L is said to be language over alphabet Σ .
- Language comprises of:
 - Set of characters – Σ
 - Set of strings (words) defined from set of character - Σ^*
 - Language L is defined from Σ^* , and $L \subseteq \Sigma^*$ because Σ^* contains many string which may not satisfy the rules of language.
- Example:
 - $\Sigma = \{a, b\}$
 - $\Sigma^* = \{\epsilon, a, b, aa, ab, ba, bb, aaa, aab, aba, abb, baa, \dots\}$

Note that:

Sets

$$\emptyset = \{ \} \neq \{ \lambda \}$$

Set size

$$|\{ \}| = |\emptyset| = 0$$

Set size

$$|\{ \lambda \}| = 1$$

String length

$$|\lambda| = 0$$

Another Example

- An infinite language $L = \{a^n b^n : n \geq 0\}$

λ
 ab
 $aabb$
 $aaaaabbbbb$

} $\in L$ $abb \notin L$

Operations on Languages

- The usual set operations

$$\{a, ab, aaaa\} \cup \{bb, ab\} = \{a, ab, bb, aaaa\}$$

$$\{a, ab, aaaa\} \cap \{bb, ab\} = \{ab\}$$

$$\{a, ab, aaaa\} - \{bb, ab\} = \{a, aaaa\}$$

- Complement:

$$\overline{L} = \Sigma^* - L$$

$$\overline{\{a, ba\}} = \{\lambda, b, aa, ab, bb, aaaa, \dots\}$$

Reverse

- Definition: $L^R = \{w^R : w \in L\}$

- Examples: $\{ab, aab, baba\}^R = \{ba, baa, abab\}$

$$L = \{a^n b^n : n \geq 0\}$$

$$L^R = \{b^n a^n : n \geq 0\}$$

Concatenation

- Definition: $L_1L_2 = \{xy : x \in L_1, y \in L_2\}$

- Example: $\{a, ab, ba\}\{b, aa\}$

$$= \{ab, aaa, abb, abaa, bab, baaa\}$$

Another Operation

- Definition:
$$L^n = \underbrace{LL \cdots L}_n$$

$$\{a,b\}^3 = \{a,b\}\{a,b\}\{a,b\} = \\ \{aaa, aab, aba, abb, baa, bab, bba, bbb\}$$

- Special case:
$$L^0 = \{\lambda\}$$

$$\{a, bba, aaa\}^0 = \{\lambda\}$$

More Examples

- $L = \{a^n b^n : n \geq 0\}$

$$L^2 = \{a^n b^n a^m b^m : n, m \geq 0\}$$

$$aabbbaaabb \in L^2$$

Star-Closure (Kleene *)

- Definition: $L^* = L^0 \cup L^1 \cup L^2 \dots$

- Example:

$$\{a, bb\}^* = \left\{ \begin{array}{l} \lambda, \\ a, bb, \\ aa, abb, bba, bbbb, \\ aaa, aabb, abba, abbbb, \dots \end{array} \right\}$$

Positive Closure

- Definition:

$$\begin{aligned} L^+ &= L^1 \cup L^2 \cup \dots \\ &= L^* - \{\lambda\} \end{aligned}$$

$$\{a, bb\}^+ = \left\{ \begin{array}{l} a, bb, \\ aa, abb, bba, bbbb, \\ aaa, aabb, abba, abbbb, \dots \end{array} \right\}$$

Computation

What is Computation?

A **computation** is any type of calculation that includes both arithmetical and non-arithmetical steps and which follows a well-defined model (e.g., an algorithm)

Computation refers to the act or action of computing like performing some CALCULATIONS. Computation refers the use or operation of a computer.

What is model?

A thing used as an example to follow or imitate.

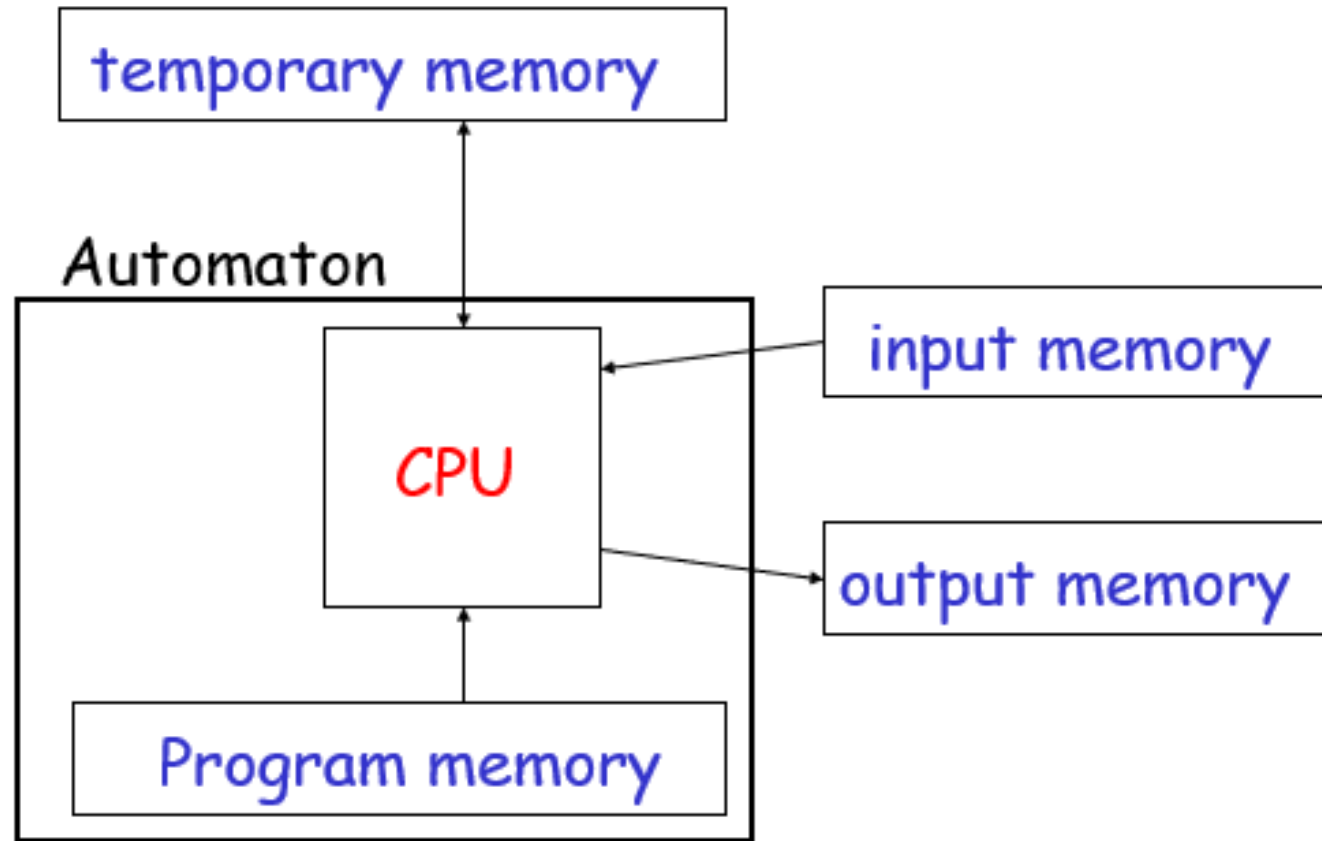
- Any Algorithm.
e.g., Algorithm for finding the factorial of any number.
- Any mathematical formula for calculating any value.
e.g., $\text{distance} = \text{speed} \times \text{time}$

Automata

Automaton

- A machine that performs a range of functions according to a predetermined set of coded instructions.
- A relatively self-operating mechanism.

Automaton

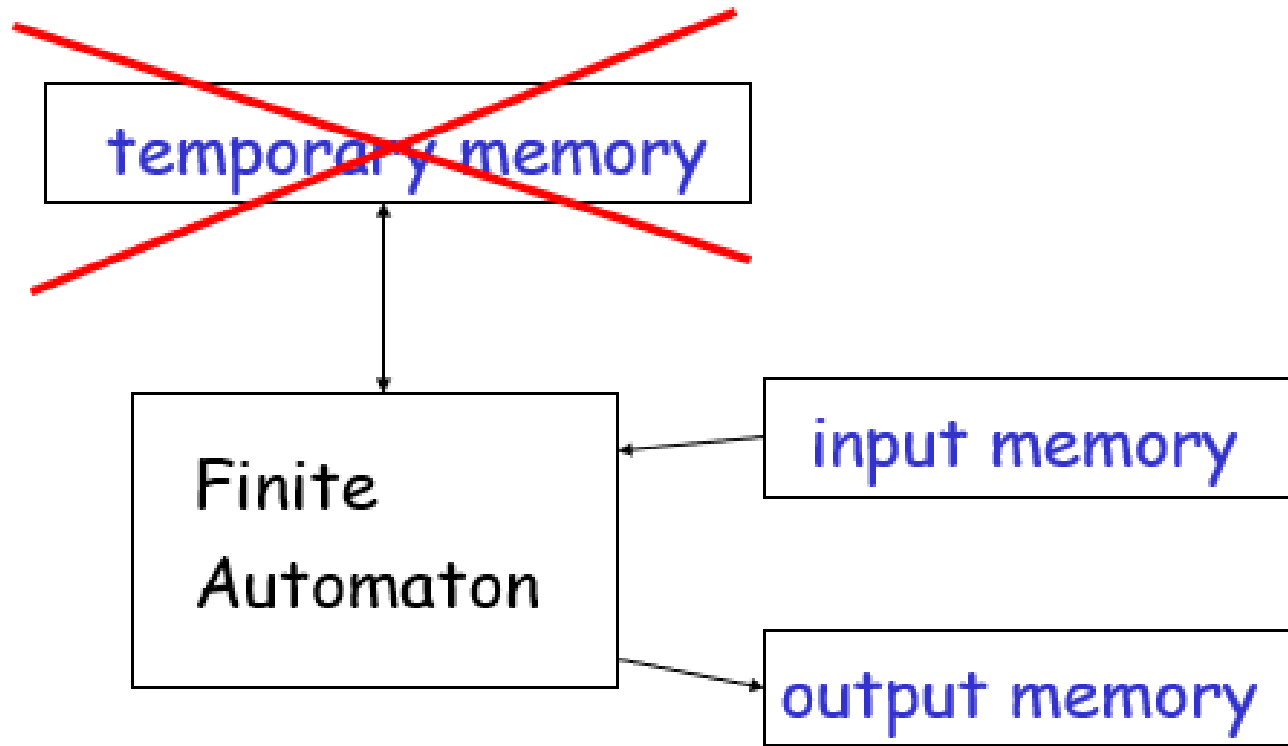


Different Kinds of Automata

Automata are distinguished by the temporary memory.

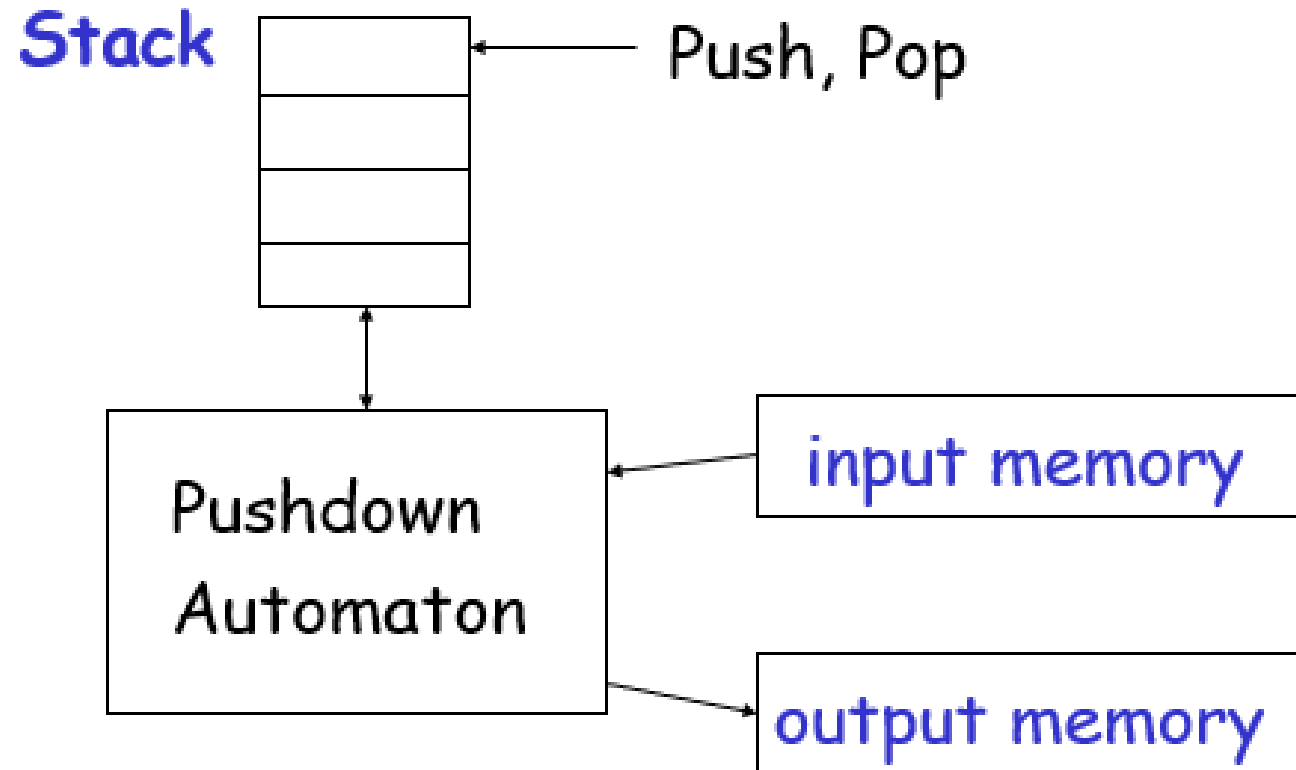
- **Finite Automata:** no temporary memory
- **Pushdown Automata:** stack
- **Turing Machines:** random access memory

Finite Automaton



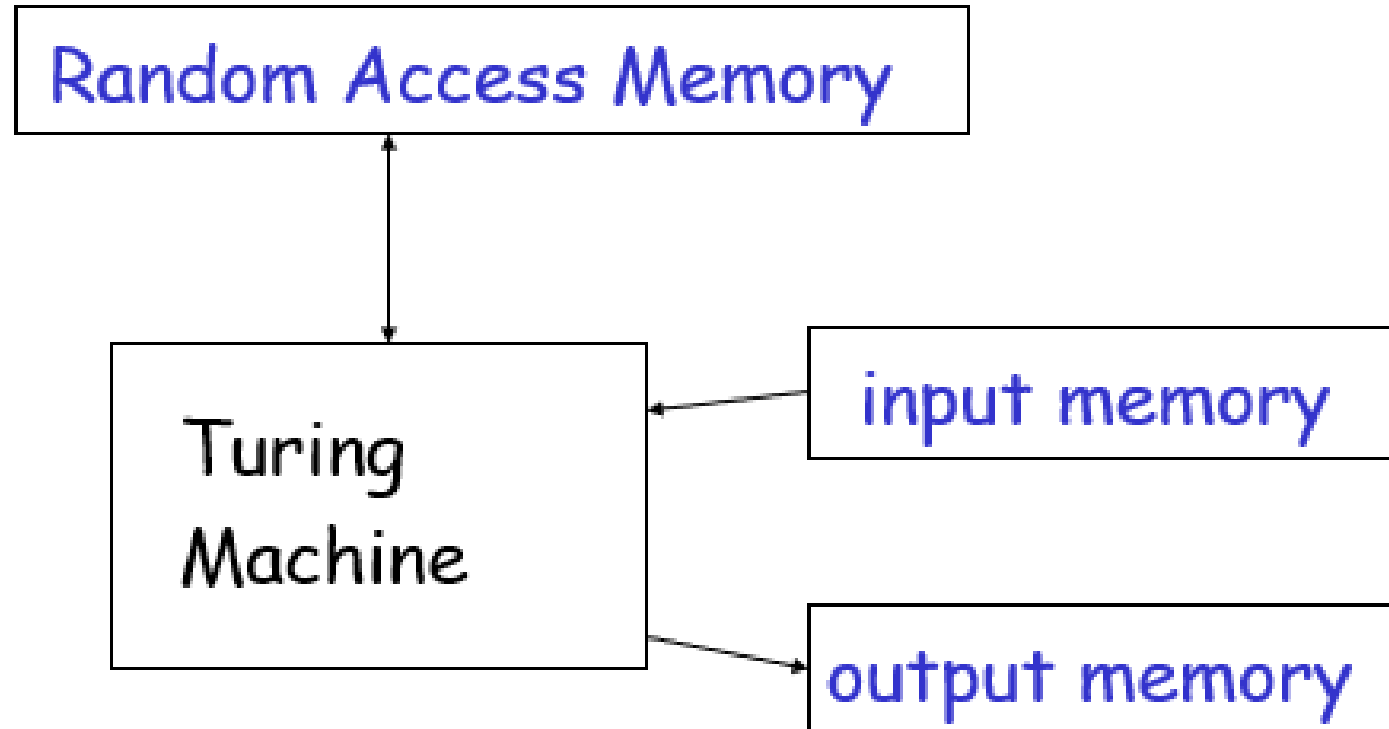
Example: Vending Machines
(small computing power)

Pushdown Automaton



Example: Compilers for Programming Languages
(medium computing power)

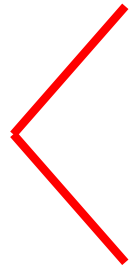
Turing Machine



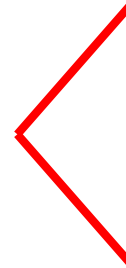
Examples: Any Algorithm
(highest computing power)

Power of Automata

Finite
Automata



Pushdown
Automata



Turing
Machine

Less power



More power

Solve more
computational problems

Finite Automata

Finite Automata

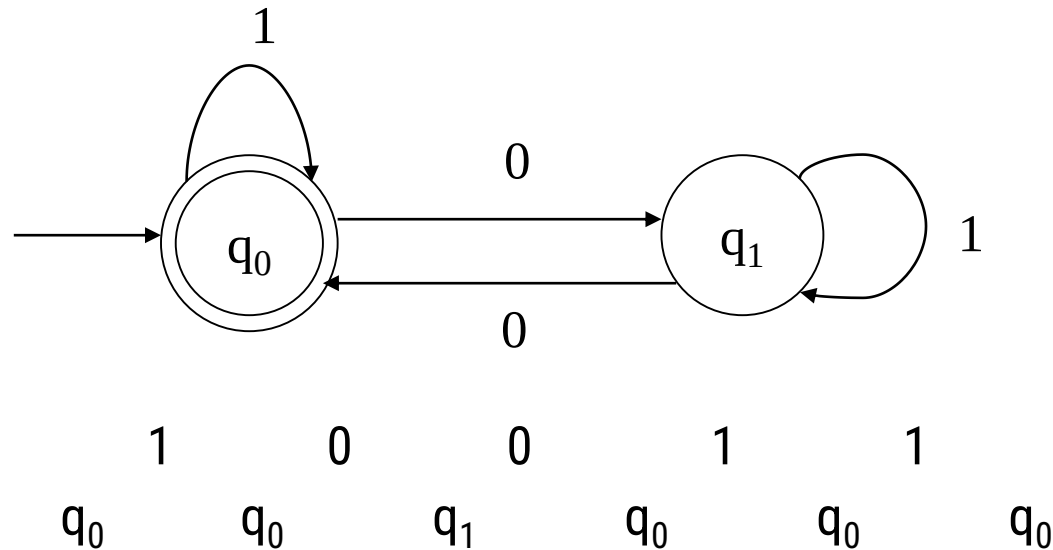
- A **finite automaton**, or **finite state machine** is a 5-tuple $(Q, \Sigma, q_0, A, \delta)$ where
 - Q is finite set of states;
 - Σ is finite alphabet of *input symbols*;
 - $q_0 \in Q$ (*initial* state);
 - $A \subseteq Q$ (the set of *accepting* states);
 - δ is a function from $Q \times \Sigma$ to Q (the *transition* function).
- For any element q of Q and any symbol $a \in \Sigma$, we interpret $\delta(q, a)$ as the state to which the FA moves, if it is in state q and receives the input a .

Applications of FA

- Lexical analysis phase of a compiler.
- String matching.

A transition diagram or table can describe the finite control:

Example #1:



One state is final/accepting, all others are rejecting.

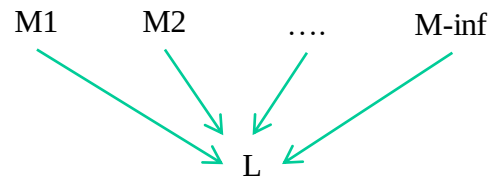
The above DFA accepts those strings that contain an even number of 0's, including the *null* string, over $\Sigma = \{0,1\}$

$L = \{\text{all strings with zero or even 0's}\}$

Note, the DFA must reject all other strings

Note

- *Machine is for accepting a language, language is the purpose!*
- *Many equivalent machines may accept the same language, but a machine cannot accept multiple languages!*

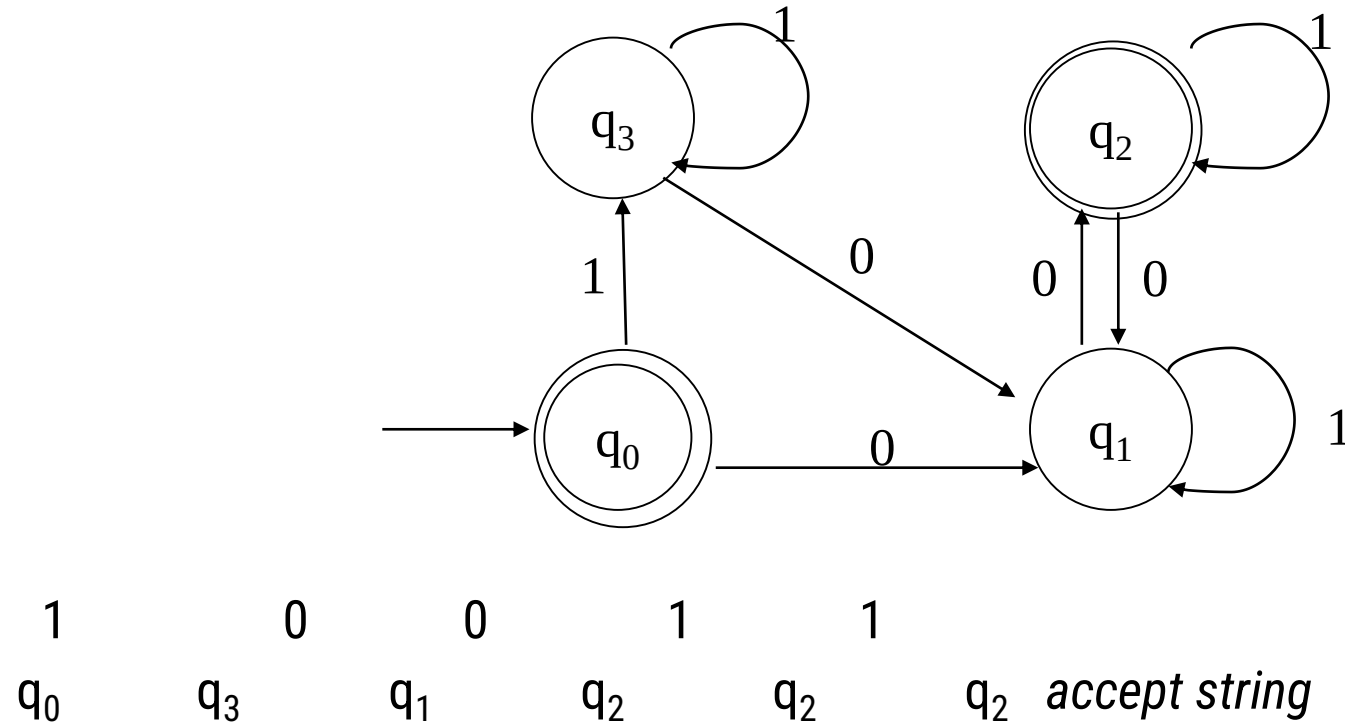


- *Id's of the characters or states are irrelevant, you can call them by any names!*

$\Sigma = \{0, 1\} \equiv \{a, b\}$

$States = \{q_0, q_1\} \equiv \{u, v\}$, as long as they have identical (isomorphic) transition table

An equivalent machine to the previous example (DFA for even number of 0's):

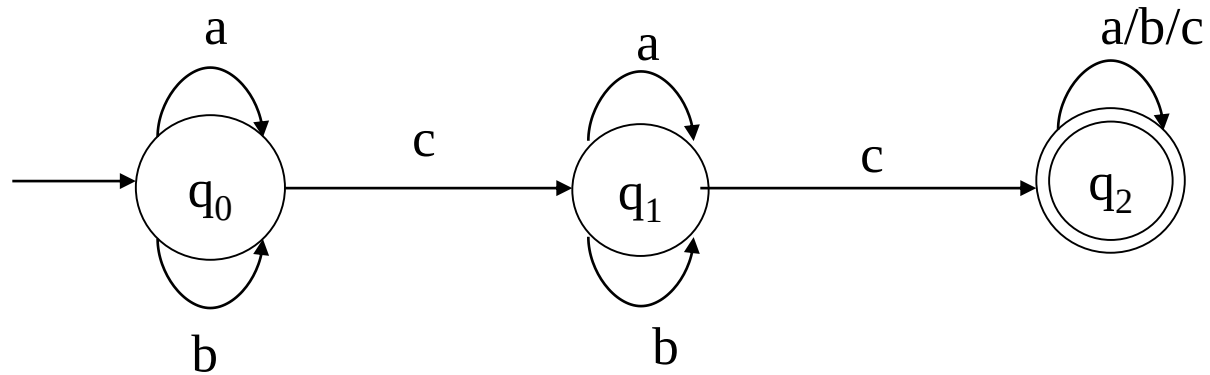


One state is final/accepting; all others are rejecting.

The above DFA accepts those strings that contain an even number of 0's, including null string, over $\Sigma = \{0,1\}$

Can you draw a machine for a language by excluding the null string from the language? $L = \{\text{all strings with 2 or more 0's}\}$

Example #2:



	a	c	c	c	b	<u>accepted</u>
q ₀	q ₀	q ₁	q ₂	q ₂	q ₂	
	a	a	c			<u>rejected</u>
q ₀	q ₀	q ₀	q ₁			

Accepts those strings that contain at least two c's

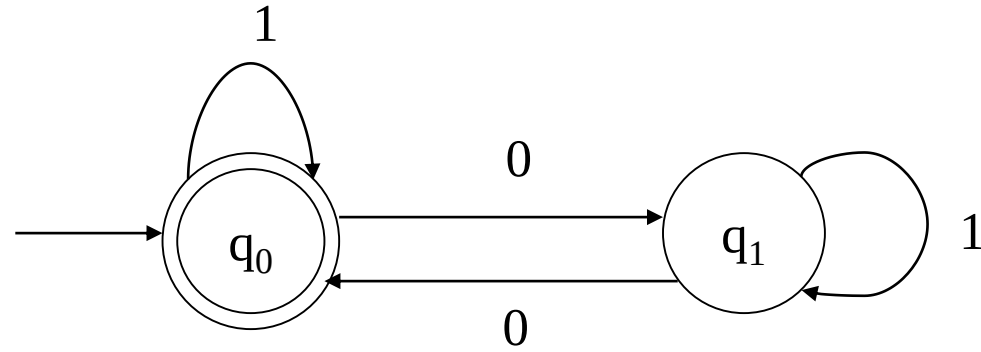
Revisit example #1:

$Q = \{q_0, q_1\}$

$\Sigma = \{0, 1\}$

Start state is q_0

$F = \{q_0\}$



δ :

	0	1
q_0	q_1	q_0
q_1	q_0	q_1

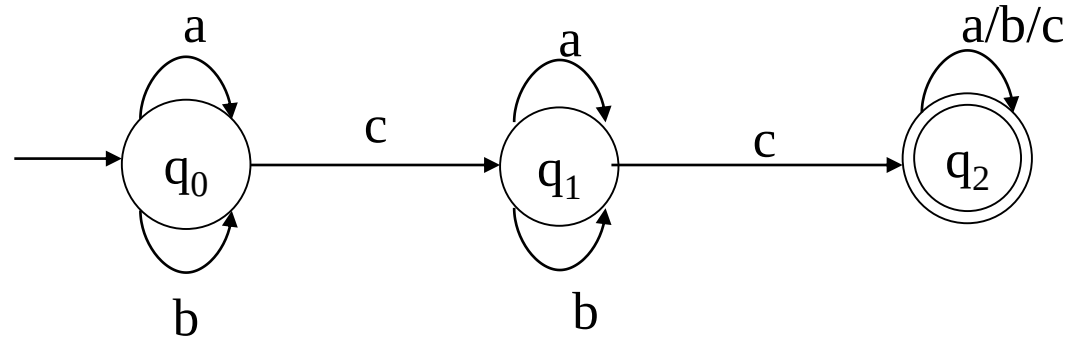
Revisit example #2:

$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b, c\}$

Start state is q_0

$F = \{q_2\}$



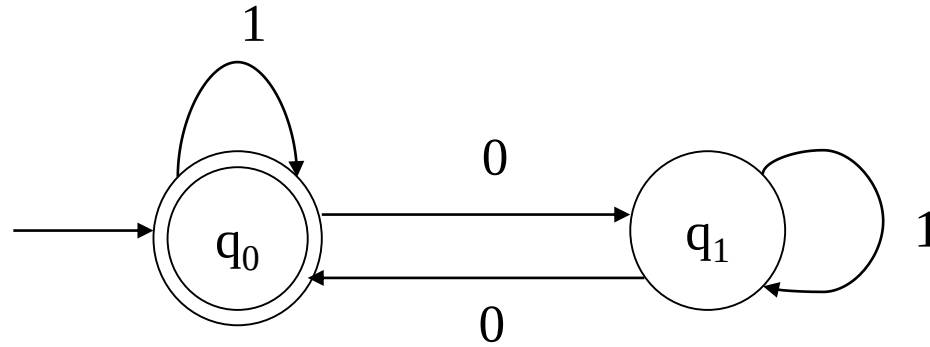
δ :

	a	b	c
q_0	q_0	q_0	q_1
q_1	q_1	q_1	q_2
q_2	q_2	q_2	q_2

Since δ is a function, at each step M has exactly one option.

It follows that for a given string, there is exactly one computation.

Recall Example #1:



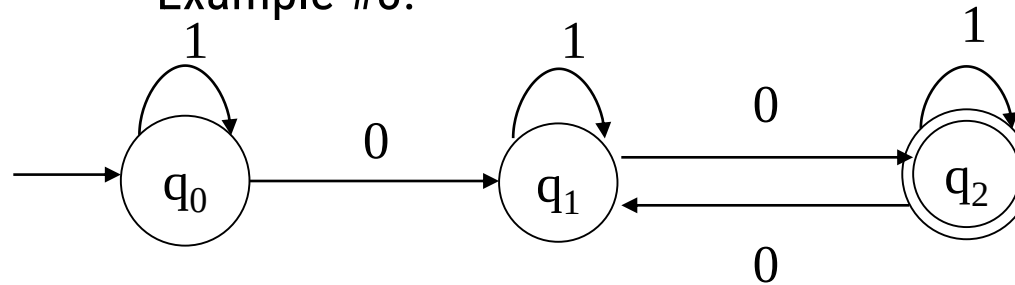
What is $\delta^*(q_0, 011)$? Informally, it is the state entered by M after processing 011 having started in state q_0 .

Formally:

$$\begin{aligned}\delta^*(q_0, 011) &= \delta(\delta^*(q_0, 01), 1) && \text{by rule \#2} \\ &= \delta(\delta(\delta^*(q_0, 0), 1), 1) && \text{by rule \#2} \\ &= \delta(\delta(\delta(\delta^*(q_0, \lambda), 0), 1), 1) && \text{by rule \#2} \\ &= \delta(\delta(\delta(q_0, 0), 1), 1) && \text{by rule \#1} \\ &= \delta(\delta(q_1, 1), 1) && \text{by definition of } \delta \\ &= \delta(q_1, 1) && \text{by definition of } \delta \\ &= q_1 && \text{by definition of } \delta\end{aligned}$$

Is 011 accepted? No, since $\delta^*(q_0, 011) = q_1$ is not a final state.

Example #3:



What is $\delta(q_0, 011)$? Informally, it is the state entered by M after processing 011 having started in state q_0 .

Formally:

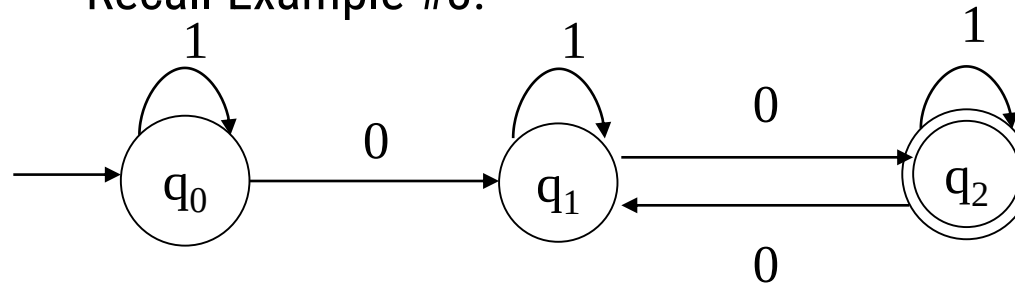
$\delta(q_0, 011)$	$= \delta(\delta(q_0, 01), 1)$	by rule #2
	$= \delta(\delta(\delta(q_0, 0), 1), 1)$	by rule #2
	$= \delta(\delta(q_1, 1), 1)$	by definition of δ
	$= \delta(q_1, 1)$	by definition of δ
	$= q_1$	by definition of δ

Is 011 accepted? No, since $\delta(q_0, 011) = q_1$ is not a final state.

Language?

$L = \{ \text{all strings over } \{0,1\} \text{ that has 2 or more } 0 \text{ symbols} \}$

Recall Example #3:



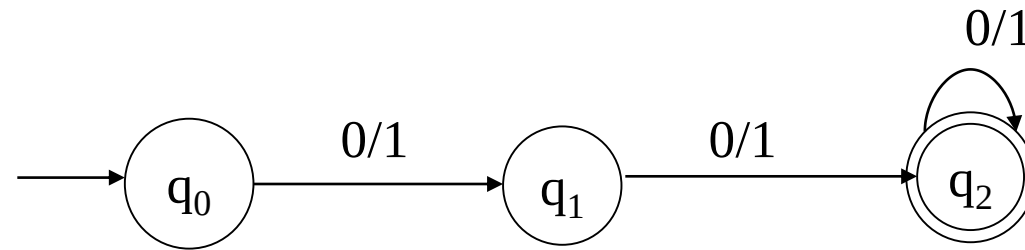
What is $\delta(q_1, 10)$?

$\delta(q_1, 10)$	$= \delta(\delta(q_1, 1), 0)$	by rule #2
	$= \delta(q_1, 0)$	by definition of δ
	$= q_2$	by definition of δ

Is 10 accepted? No, since $\delta(q_0, 10) = q_1$ is not a final state. The fact that $\delta(q_1, 10) = q_2$ is irrelevant, q_1 is not the start state!

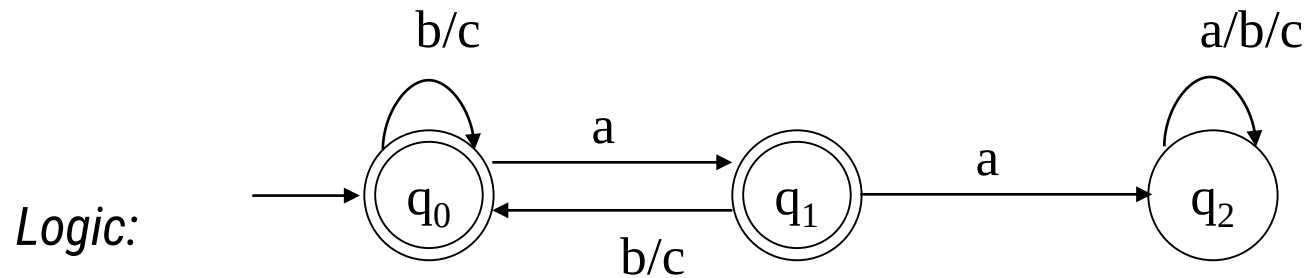
Give a DFA M such that:

$$L(M) = \{x \mid x \text{ is a string of 0's and 1's and } |x| \geq 2\}$$



Give a DFA M such that:

$L(M) = \{x \mid x \text{ is a string of (zero or more) a's, b's, and c's such that } x \text{ does not contain the substring } aa\}$



*In Start state (q_0): b's and c's: ignore – stay in the same state
 q_0 is also an “accept” state*

First ‘a’ appears: get ready (q_1) to reject

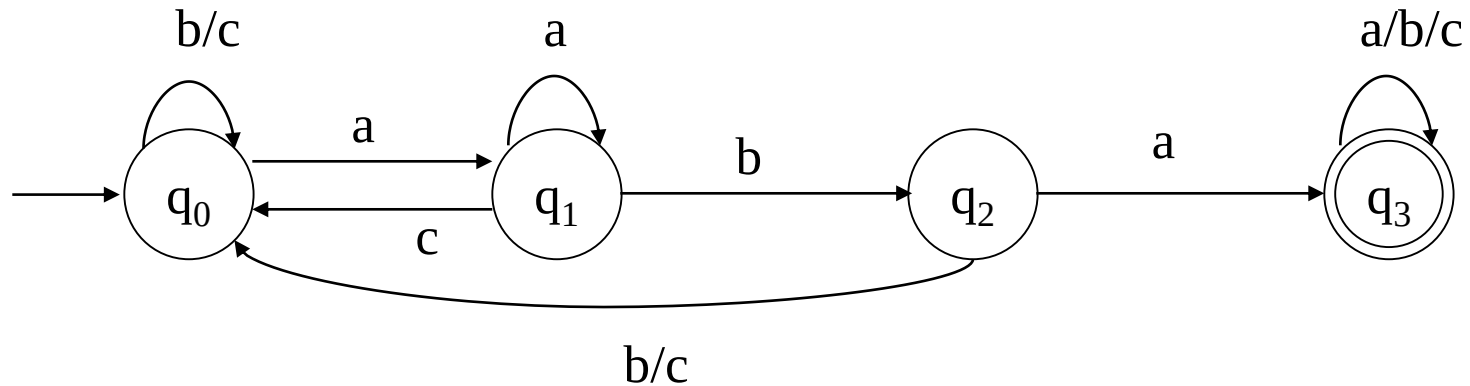
But followed by a ‘b’ or ‘c’: go back to start state q_0

When the second ‘a’ appears after the “ready” state, go to reject state q_2

Ignore everything after getting to the “reject” state q_2

Give a DFA M such that:

$L(M) = \{x \mid x \text{ is a string of a's, b's and c's such that } x \text{ contains the substring } aba\}$



Logic: acceptance is straightforward, progressing on each expected symbol

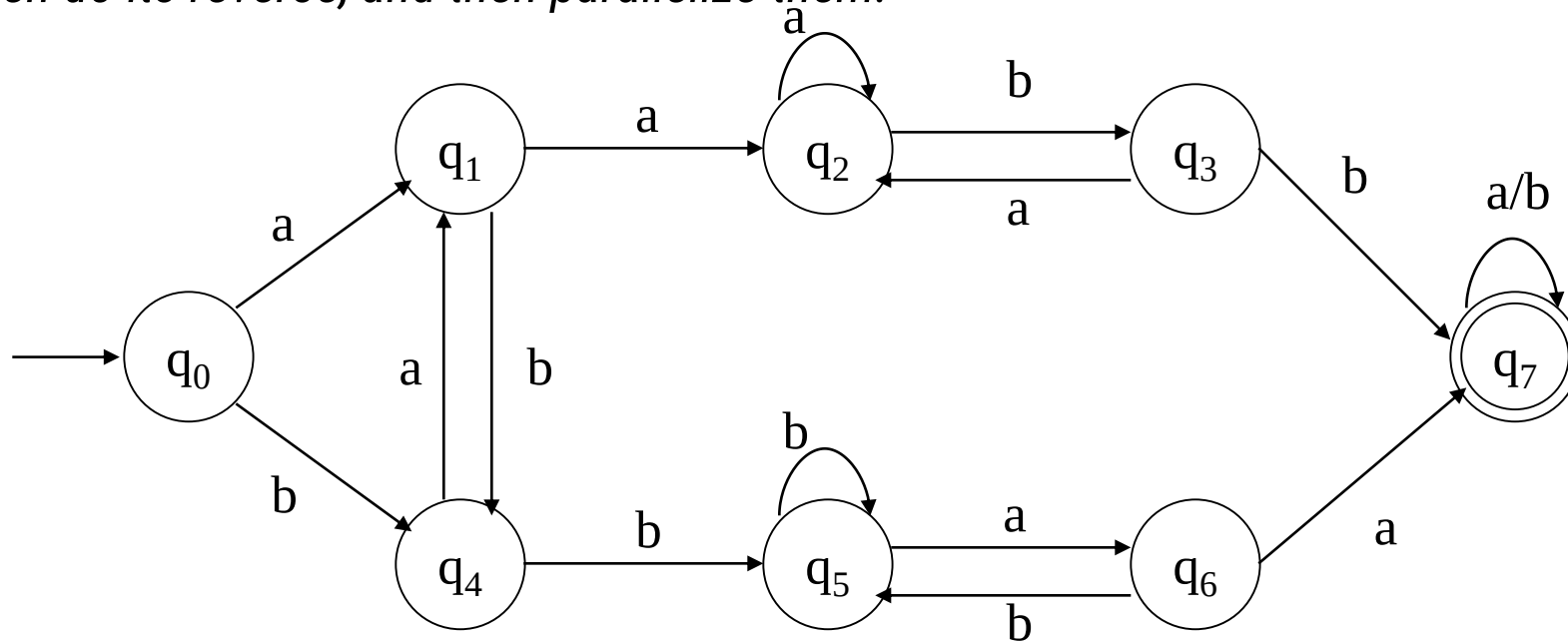
However, rejection needs special care in each state (for DFA, we will see this becomes easier in NFA, a non-deterministic machine)

Give a DFA M such that:

$L(M) = \{x \mid x \text{ is a string of a's and b's such that } x \text{ contains both } aa \text{ and } bb\}$

First do, for a language where 'aa' comes before 'bb'

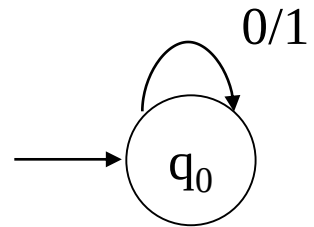
Then do its reverse, and then parallelize them.



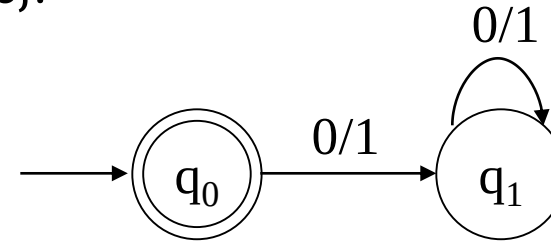
Remember, you may have multiple "final" states but only one "start" state.

Let $\Sigma = \{0, 1\}$. Give DFAs for $\{\}$, $\{\epsilon\}$, Σ^* , and Σ^+ .

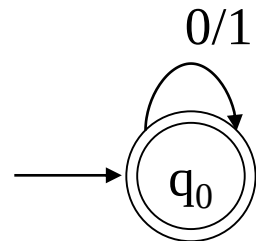
For $\{\}$:



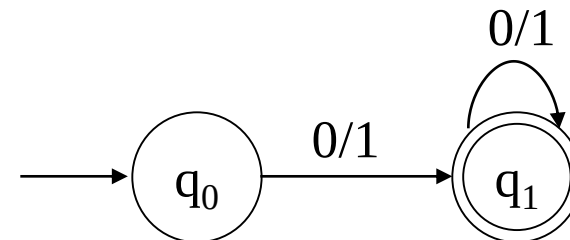
For $\{\epsilon\}$:



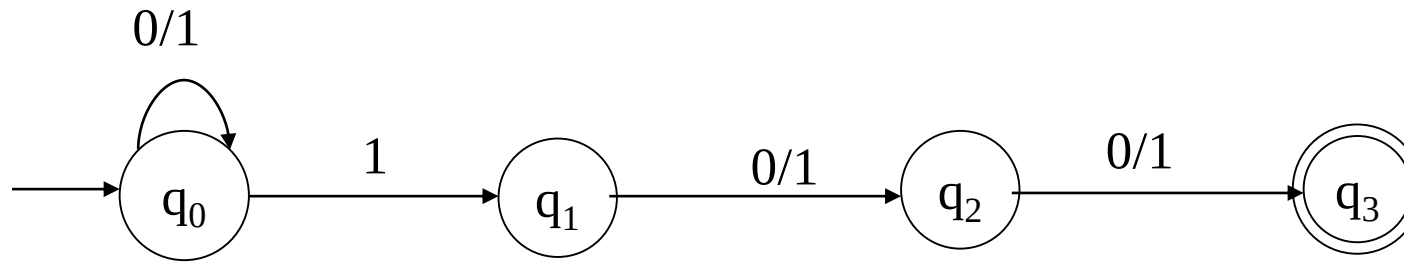
For Σ^* :



For Σ^+ :



Problem: Third symbol from last is 1



Is this a DFA?

No, but it is a *Non-deterministic Finite Automaton*

Nondeterministic Finite State Automata (NFA)

An NFA is a five-tuple:

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q A finite set of states

Σ A finite input alphabet

q_0 The initial/starting state, q_0 is in Q

F A set of final/accepting states, which is a subset of Q

δ A transition function, which is a total function from $Q \times \Sigma$ to 2^Q

$\delta: (Q \times \Sigma) \rightarrow 2^Q$: 2^Q is the power set of Q , the set of *all subsets* of Q

$\delta(q,s)$: The set of all states p such that there is a transition labeled s from q to p

$\delta(q,s)$ is a function from $Q \times S$ to 2^Q (but not only to Q)

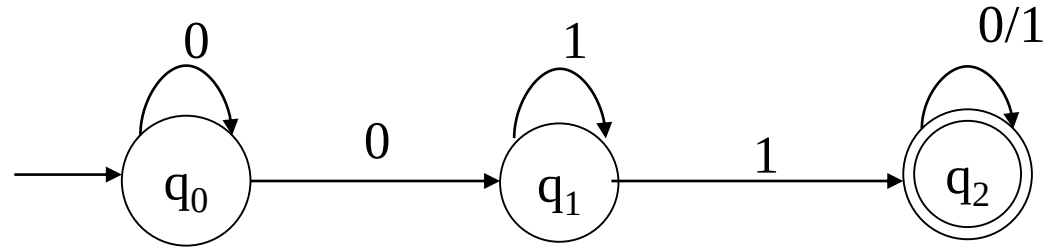
Example #1: one or more 0's followed by one or more 1's

$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{0, 1\}$

Start state is q_0

$F = \{q_2\}$



δ :

	0	1
q_0	$\{q_0, q_1\}$	$\{\}$
q_1	$\{\}$	$\{q_1, q_2\}$
q_2	$\{q_2\}$	$\{q_2\}$

Example #2: pair of 0's or pair of 1's as substring

$Q = \{q_0, q_1, q_2, q_3, q_4\}$

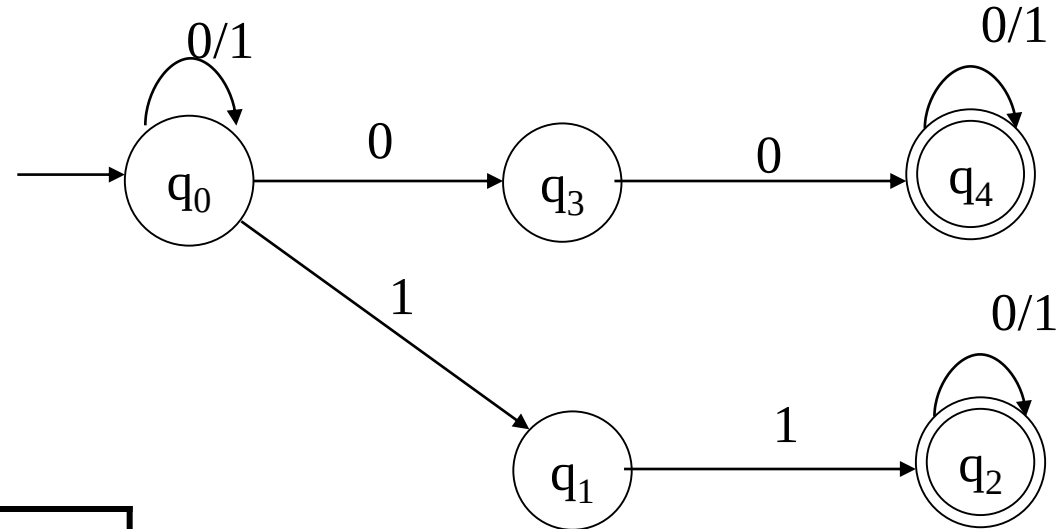
$\Sigma = \{0, 1\}$

Start state is q_0

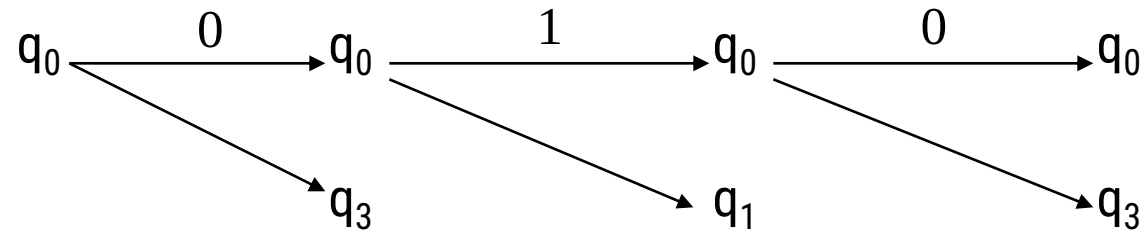
$F = \{q_2, q_4\}$

δ :

	0	1
q_0	$\{q_0, q_3\}$	$\{q_0, q_1\}$
q_1	$\{\}$	$\{q_2\}$
q_2	$\{q_2\}$	$\{q_2\}$
q_3	$\{q_4\}$	$\{\}$
q_4	$\{q_4\}$	$\{q_4\}$

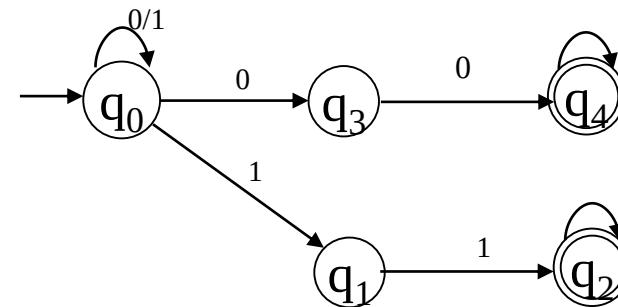


Another example (010):



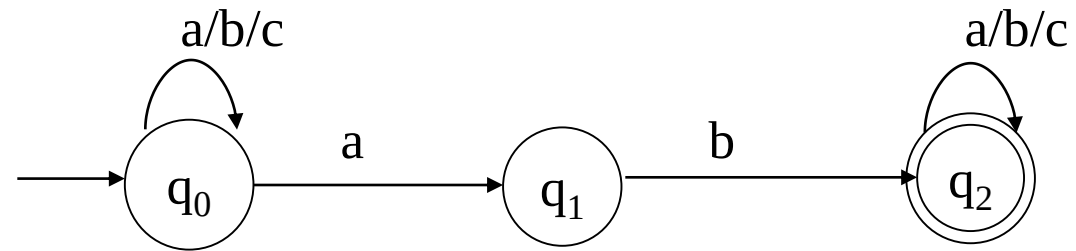
not accepted

All paths have been explored, and none lead to an accepting state.



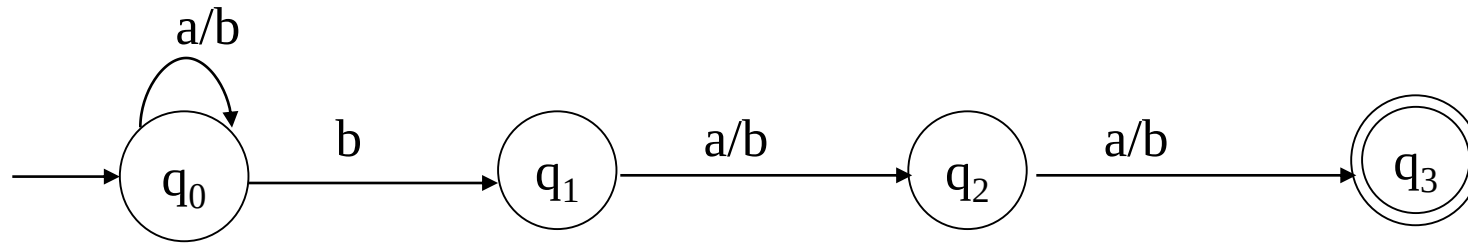
Let $\Sigma = \{a, b, c\}$. Give an NFA M that accepts:

$L = \{x \mid x \text{ is in } \Sigma^* \text{ and } x \text{ contains } ab\}$



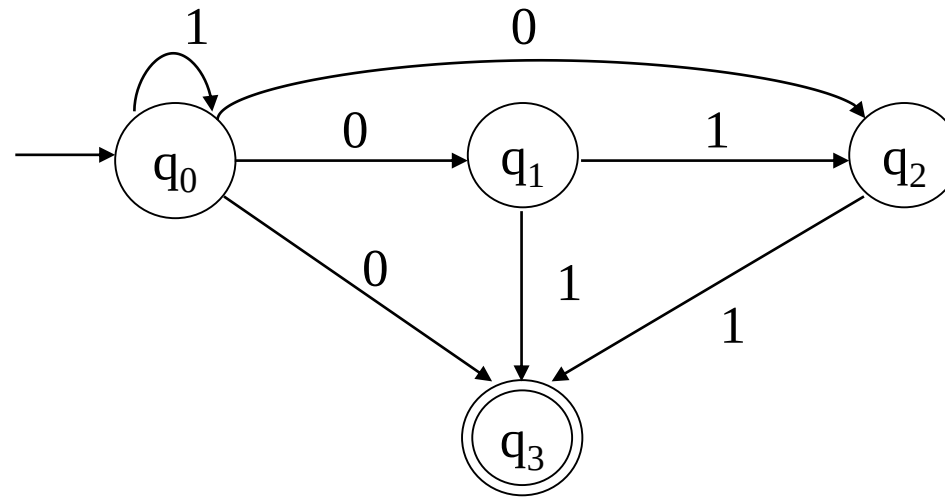
Let $\Sigma = \{a, b\}$. Give an NFA M that accepts:

$L = \{x \mid x \text{ is in } \Sigma^* \text{ and the third to the last symbol in } x \text{ is } b\}$



Give an equivalent DFA as an exercise.

Example:



What is $\delta(\{q_0\}, 10)$?

Informally: The set of states the NFA could be in after processing 10, having started in state q_0 , i.e., $\{q_1, q_2, q_3\}$.

Formally:

$$\begin{aligned}\delta(\{q_0\}, 10) &= \delta(\delta(\{q_0\}, 1), 0) \\ &= \delta(\{q_0\}, 0) \\ &= \{q_1, q_2, q_3\}\end{aligned}$$

Is 10 accepted? Yes!

Example:

What is $\delta(\{q_0, q_1\}, 1)$?

$$\begin{aligned}\delta(\{q_0, q_1\}, 1) &= \delta(\{q_0\}, 1) \cup \delta(\{q_1\}, 1) \\ &= \{q_0\} \cup \{q_2, q_3\} \\ &= \{q_0, q_2, q_3\}\end{aligned}$$

What is $\delta(\{q_0, q_2\}, 10)$?

$$\begin{aligned}\delta(\{q_0, q_2\}, 10) &= \delta(\delta(\{q_0, q_2\}, 1), 0) \\ &= \delta(\delta(\{q_0\}, 1) \cup \delta(\{q_2\}, 1), 0) \\ &= \delta(\{q_0\} \cup \{q_3\}, 0) \\ &= \delta(\{q_0, q_3\}, 0) \\ &= \delta(\{q_0\}, 0) \cup \delta(\{q_3\}, 0) \\ &= \{q_1, q_2, q_3\} \cup \{\} \\ &= \{q_1, q_2, q_3\}\end{aligned}$$

Example:

$$\begin{aligned}\delta(\{q_0\}, 101) &= \delta(\delta(\{q_0\}, 10), 1) \\ &= \delta(\delta(\delta(\{q_0\}, 1), 0), 1) \\ &= \delta(\delta(\{q_0\}, 0), 1) \\ &= \delta(\{q_1, q_2, q_3\}, 1) \\ &= \delta(\{q_1\}, 1) \cup \delta(\{q_2\}, 1) \cup \delta(\{q_3\}, 1) \\ &= \{q_2, q_3\} \cup \{q_3\} \cup \{\} \\ &= \{q_2, q_3\}\end{aligned}$$

Is 101 accepted? Yes! q_3 is a final state.

NFAs with ϵ Moves

An NFA- ϵ is a five-tuple:

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q A finite set of states

Σ A finite input alphabet

q_0 The initial/starting state, q_0 is in Q

F A set of final/accepting states, which is a subset of Q

δ A transition function, which is a total function from $Q \times \Sigma \cup \{\epsilon\}$ to 2^Q

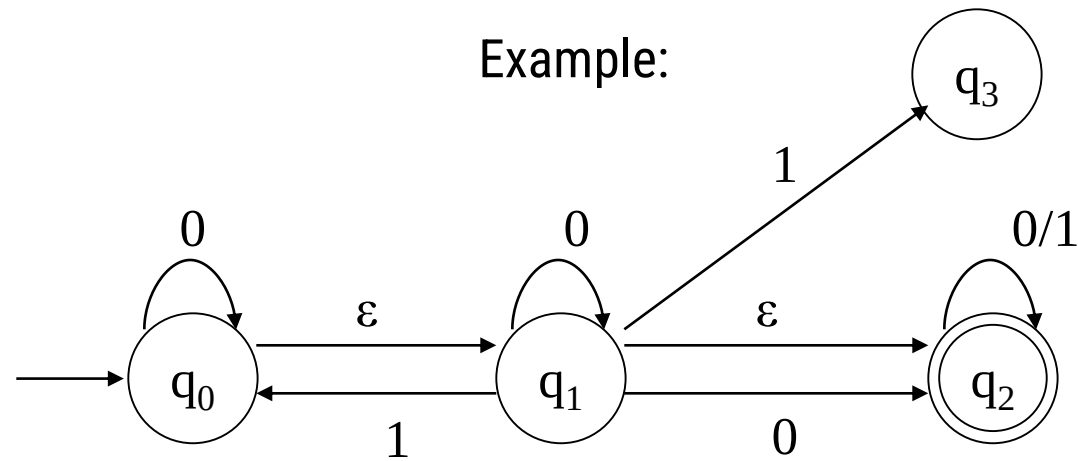
$$\delta: (Q \times (\Sigma \cup \{\epsilon\})) \rightarrow 2^Q$$

$$\delta(q, s)$$

-The set of all states p such that there is a transition labeled a from q to p , where a is in $\Sigma \cup \{\epsilon\}$

Sometimes referred to as an NFA- ϵ other times, simply as an NFA.

Example:



δ :

	0	1	ϵ
q_0	$\{q_0\}$	$\{ \}$	$\{q_1\}$
q_1	$\{q_1, q_2\}$	$\{q_0, q_3\}$	$\{q_2\}$
q_2	$\{q_2\}$	$\{q_2\}$	$\{ \}$
q_3	$\{ \}$	$\{ \}$	$\{ \}$

A string $w = w_1 w_2 \dots w_n$ is processed
as $w = \epsilon^* w_1 \epsilon^* w_2 \epsilon^* \dots \epsilon^* w_n \epsilon^*$

Example: all computations on 00:

$0 \quad \epsilon \quad 0$
 $q_0 \quad q_0 \quad q_1 \quad q_2$

ϵ -closure

Define ϵ -closure(q) to denote the set of all states reachable from q by zero or more ϵ transitions.

Examples: (for the previous NFA)

$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\} \quad \epsilon\text{-closure}(q_2) = \{q_2\}$$

$$\epsilon\text{-closure}(q_1) = \{q_1, q_2\} \quad \epsilon\text{-closure}(q_3) = \{q_3\}$$

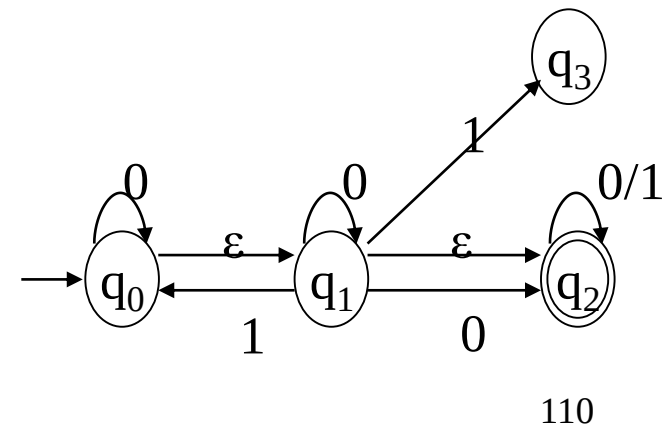
ϵ -closure(q) can be extended to sets of states by defining:

$$\epsilon\text{-closure}(P) = \bigcup_{q \in P} \epsilon\text{-closure}(q)$$

Examples:

$$\epsilon\text{-closure}(\{q_1, q_2\}) = \{q_1, q_2\}$$

$$\epsilon\text{-closure}(\{q_0, q_3\}) = \{q_0, q_1, q_2, q_3\}$$



Minimization of FA

Example: Minimize FA

Final state is {6}

And, Non-Final state is {1,2,3,4,5,7}

(6, 1), (6,2), (6, 3), (6, 4), (6, 5), (6, 7) are distinguish pairs.

Consider pair **(1,2)**

$$\delta(1,a)=2$$

$$\delta(1,b)=3$$

$$\delta(2,a)=4$$

$$\delta(2,b)=5$$

Consider pair **(1,3)**

$$\delta(1,a)=2$$

$$\delta(1,b)=3$$

$$\delta(3,a)=6$$

$$\delta(3,b)=7$$

pair (2,6) is distinguish, so (1,3) is distinguished pair.

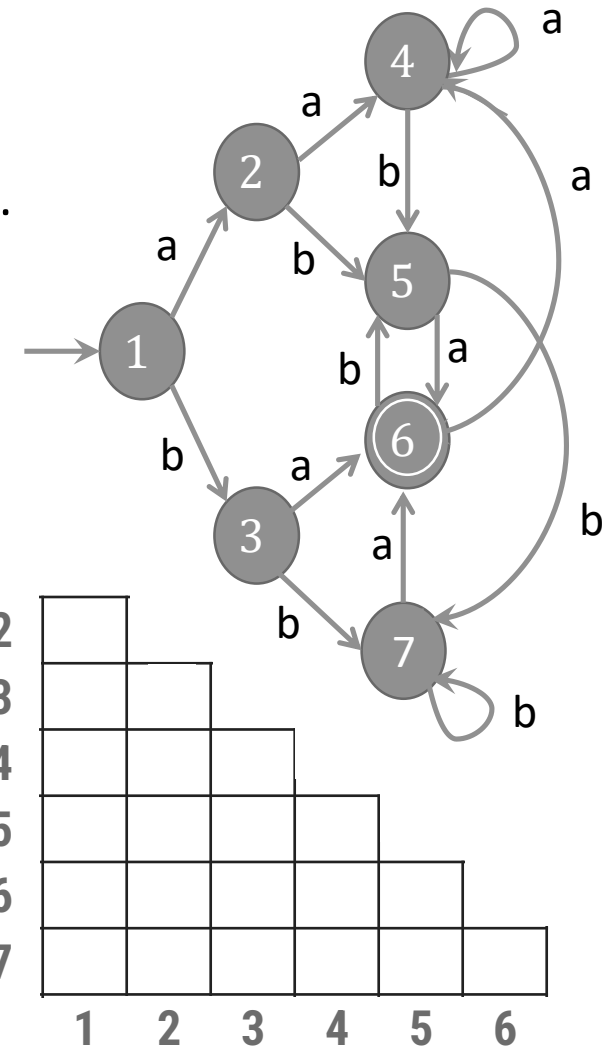
Consider pair **(1,4)**

$$\delta(1,a)=2$$

$$\delta(1,b)=3$$

$$\delta(4,a)=4$$

$$\delta(4,b)=5$$



Example: Minimize FA

Consider pair **(1,5)**

$$\delta(1,a)=2$$

$$\delta(1,b)=3$$

$$\delta(5,a)=6$$

$$\delta(5,b)=7$$

pair (2,6) is distinguish, so (1,5) is distinguished pair.

Consider pair **(1,7)**

$$\delta(1,a)=2$$

$$\delta(1,b)=3$$

$$\delta(7,a)=6$$

$$\delta(7,b)=7$$

pair (2,6) is distinguish, so (1, 7) is distinguished pair.

Consider pair **(2,3)**

$$\delta(2,a)=4$$

$$\delta(2,b)=5$$

$$\delta(3,a)=6$$

$$\delta(3,b)=7$$

pair (4,6) is distinguish, so (2, 3) distinguished pair.

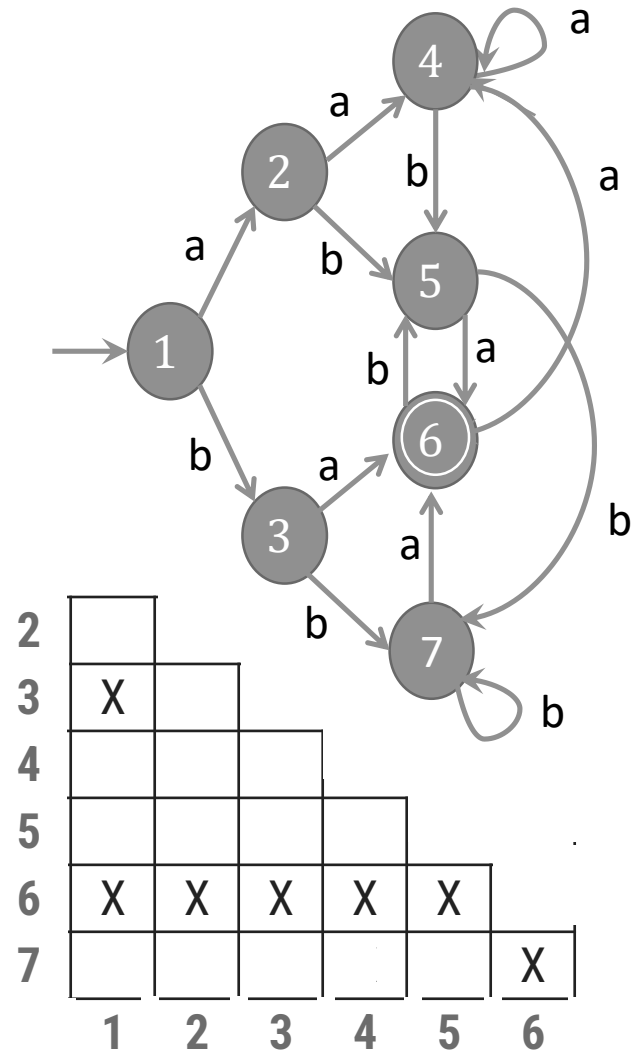
Consider pair **(2,4)**

$$\delta(2,a)=4$$

$$\delta(2,b)=5$$

$$\delta(4,a)=4$$

$$\delta(4,b)=5$$



Example: Minimize FA

Consider pair **(2,5)**

$$\delta(2,a)=4$$

$$\delta(2,b)=5$$

$$\delta(5,a)=6$$

$$\delta(5,b)=7$$

pair (4,6) is distinguish, so (2,5) is distinguished pair.

Consider pair **(2,7)**

$$\delta(2,a)=4$$

$$\delta(2,b)=5$$

$$\delta(7,a)=6$$

$$\delta(7,b)=7$$

pair (4,6) is distinguish, so (2,7) is distinguished pair.

Consider pair **(3,4)**

$$\delta(3,a)=6$$

$$\delta(3,b)=7$$

$$\delta(4,a)=4$$

$$\delta(4,b)=5$$

pair (6,4) is distinguish, so (3,4) is distinguish pair.

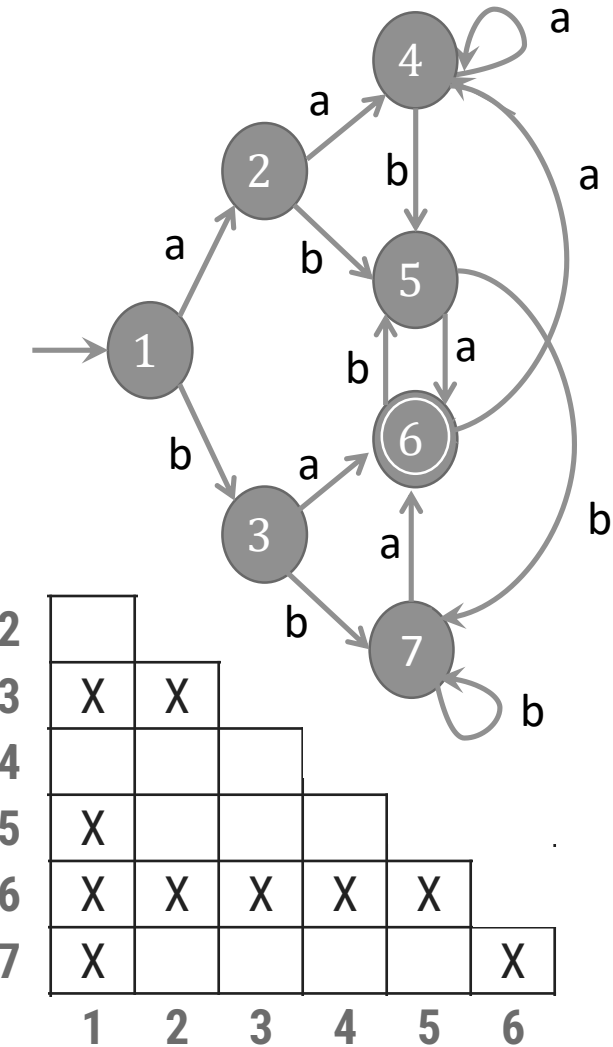
Consider pair **(3,5)**

$$\delta(3,a)=6$$

$$\delta(3,b)=7$$

$$\delta(5,a)=6$$

$$\delta(5,b)=7$$



Example: Minimize FA

Consider pair **(3,7)**

$$\delta(3,a)=6$$

$$\delta(3,b)=7$$

$$\delta(7,a)=6$$

$$\delta(7,b)=7$$

Consider pair **(4,5)**

$$\delta(4,a)=4$$

$$\delta(4,b)=5$$

$$\delta(5,a)=6$$

$$\delta(5,b)=7$$

pair (6,4) is distinguish, so (4,5) is distinguish.

Consider pair **(4,7)**

$$\delta(4,a)=4$$

$$\delta(4,b)=5$$

$$\delta(7,a)=6$$

$$\delta(7,b)=7$$

pair (4,6) is distinguish, so (4,7) is distinguish.

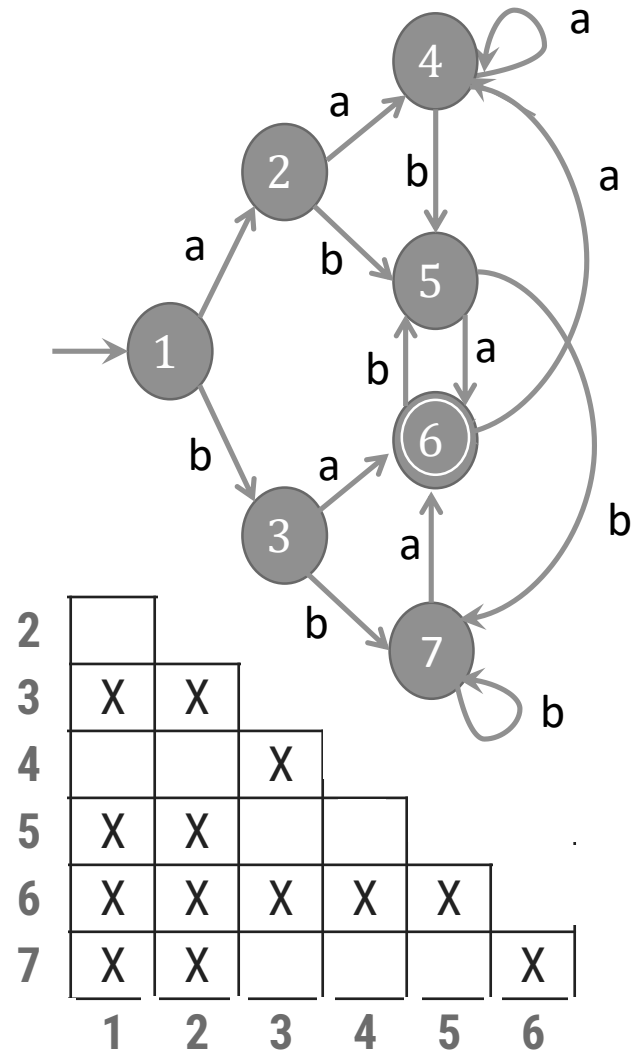
Consider pair **(5,7)**

$$\delta(5,a)=6$$

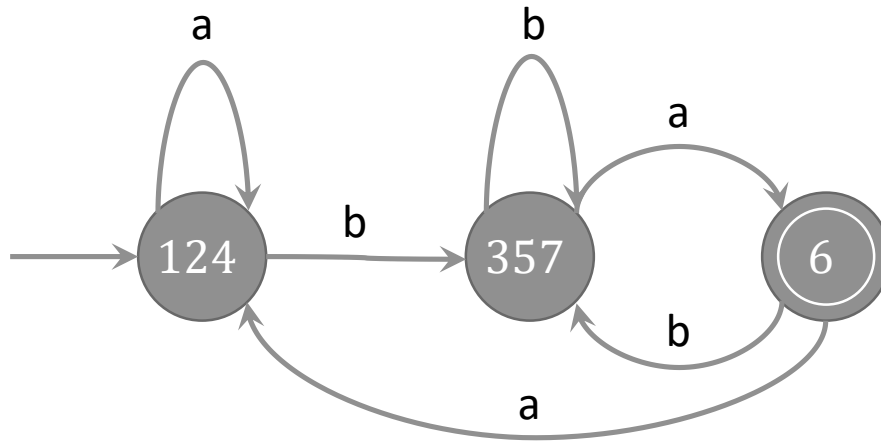
$$\delta(5,b)=7$$

$$\delta(7,a)=6$$

$$\delta(7,b)=7$$

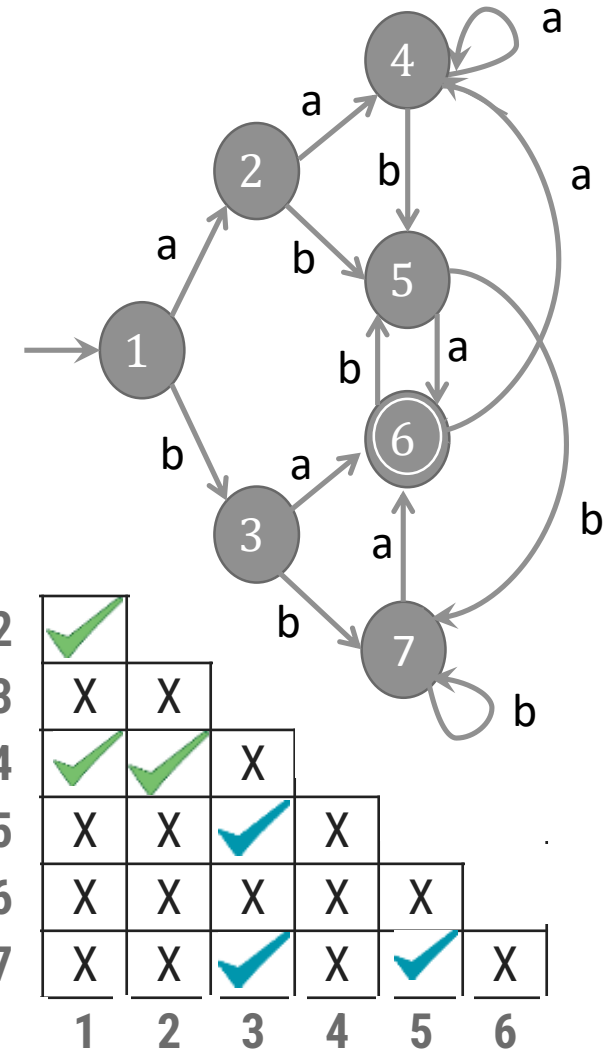


Example: Minimize FA



Minimized FA

1=2 1=4 2=4 $\longrightarrow$ 1=2=4
3=5 3=7 5=7 $\longrightarrow$ 3=5=7



Thank You